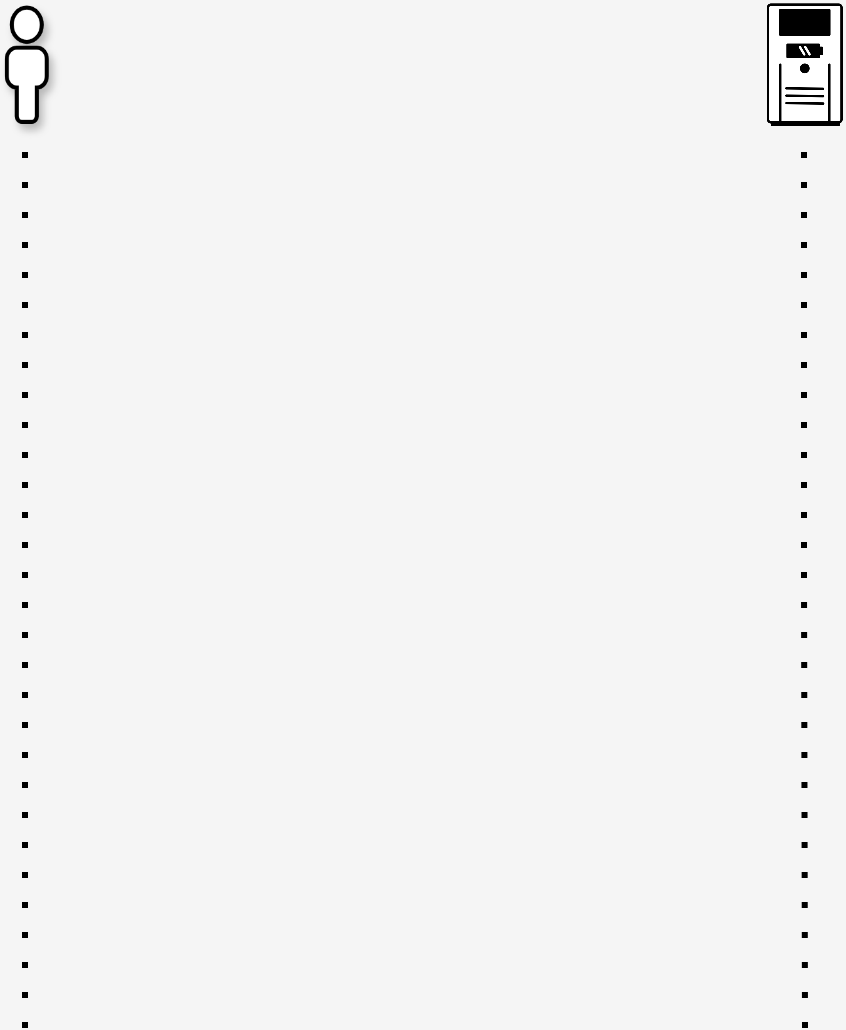
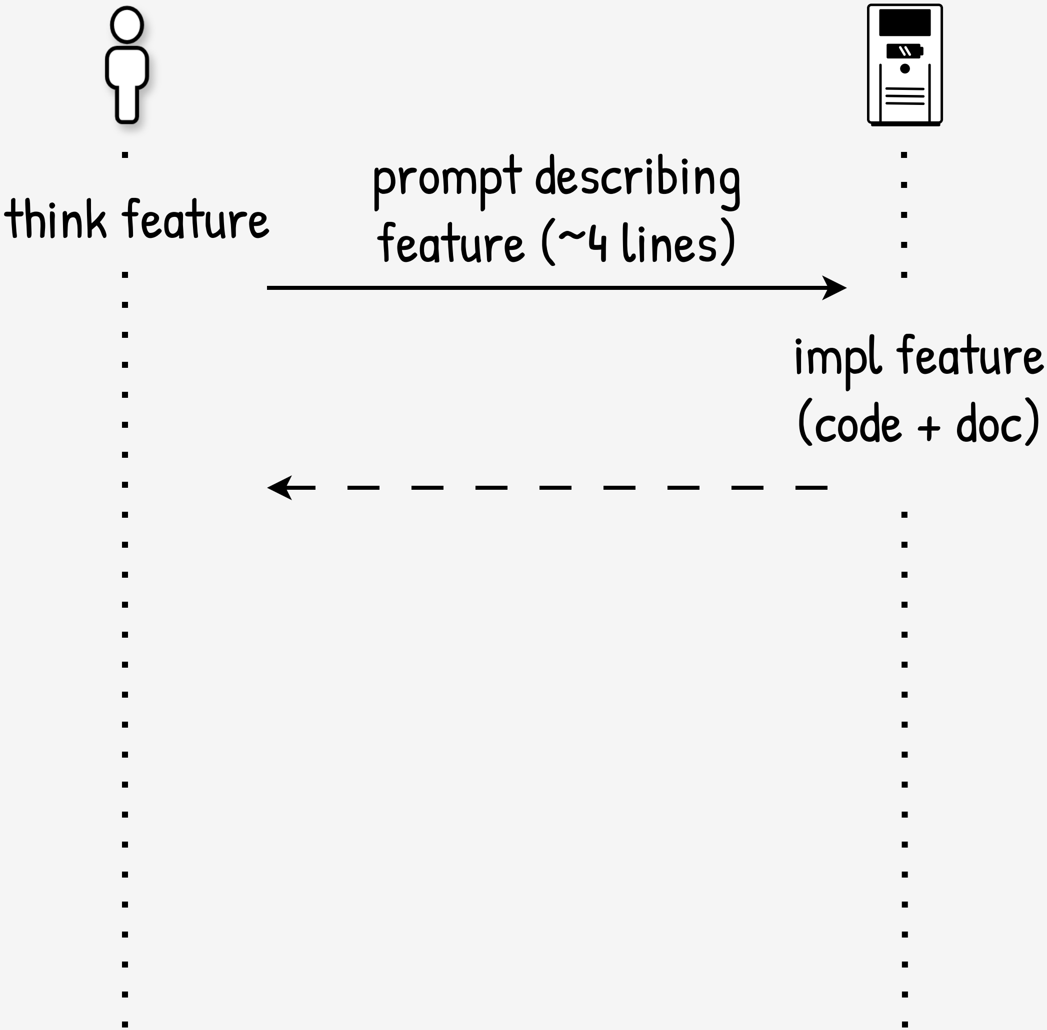
Using AI for Software Developing

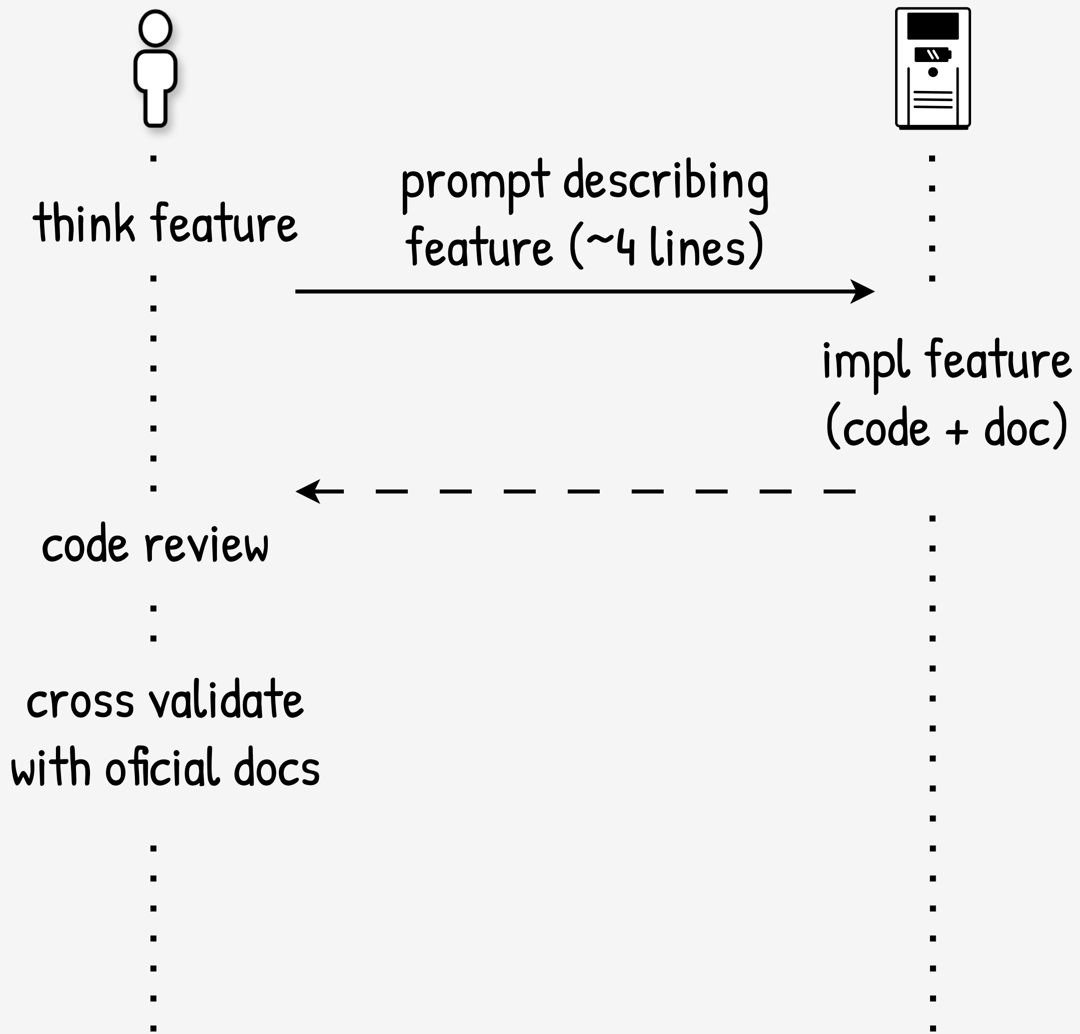
(a personal journey)

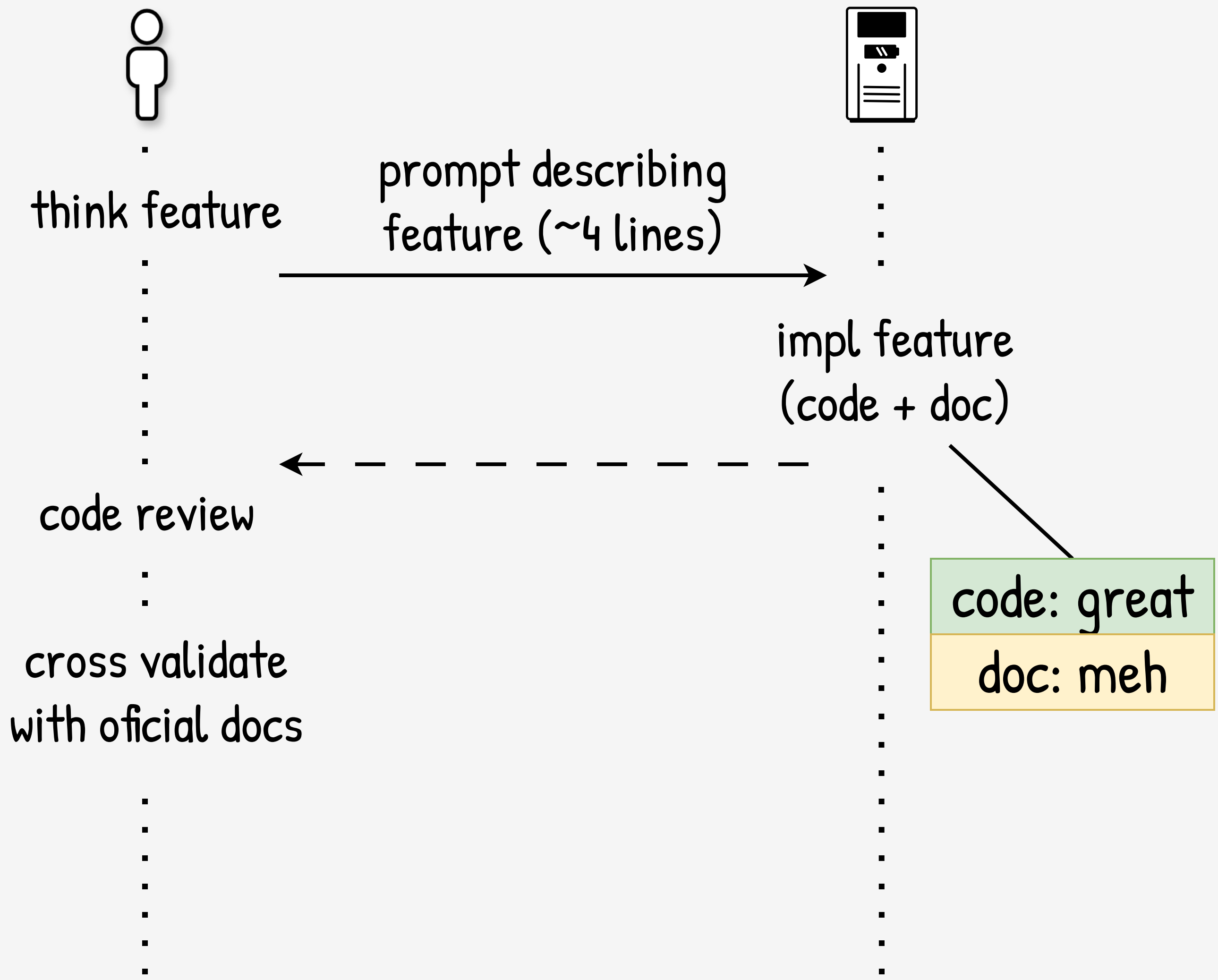
Taller de Programación

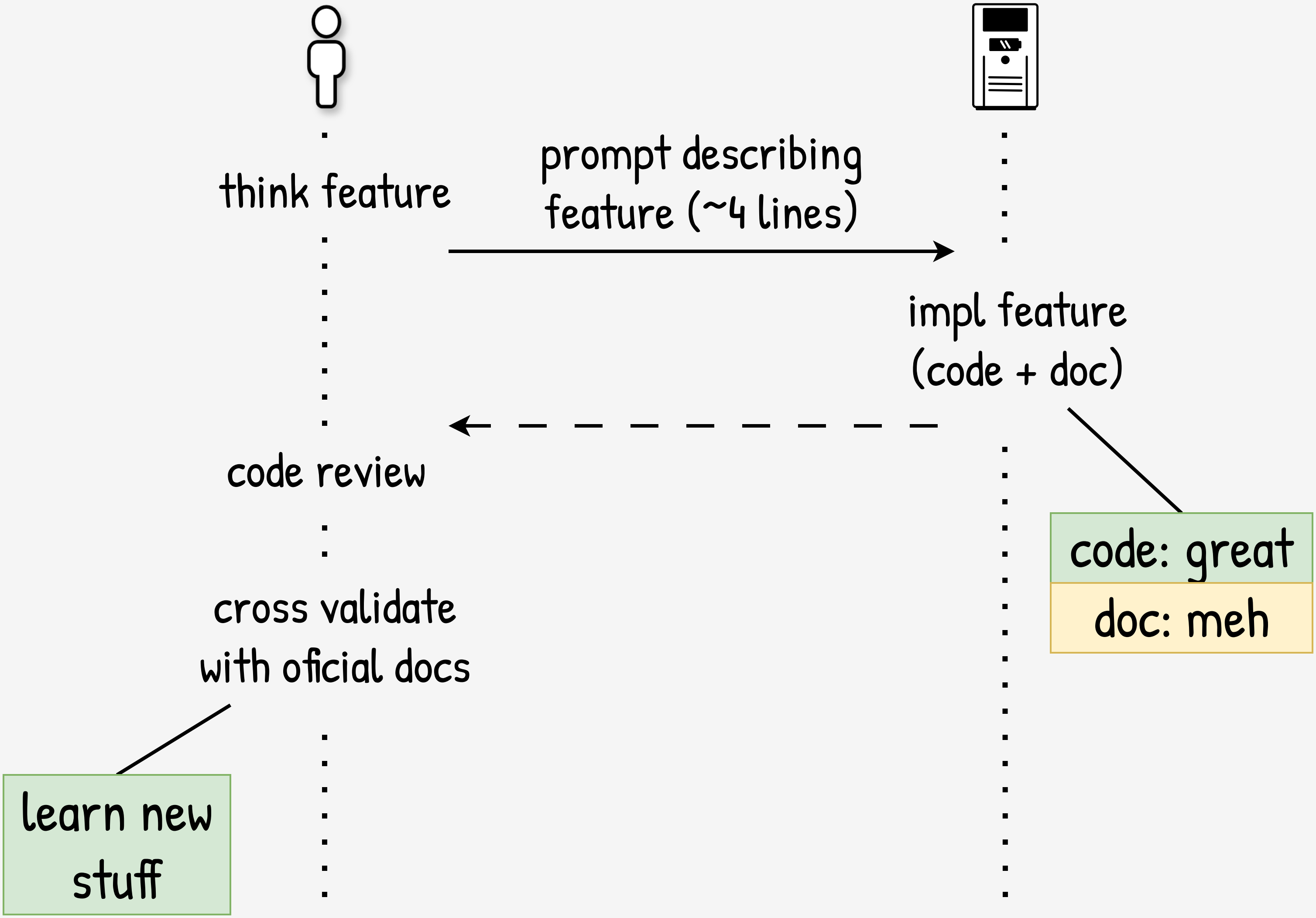
Ing. Martin Di Paola - Fiuba

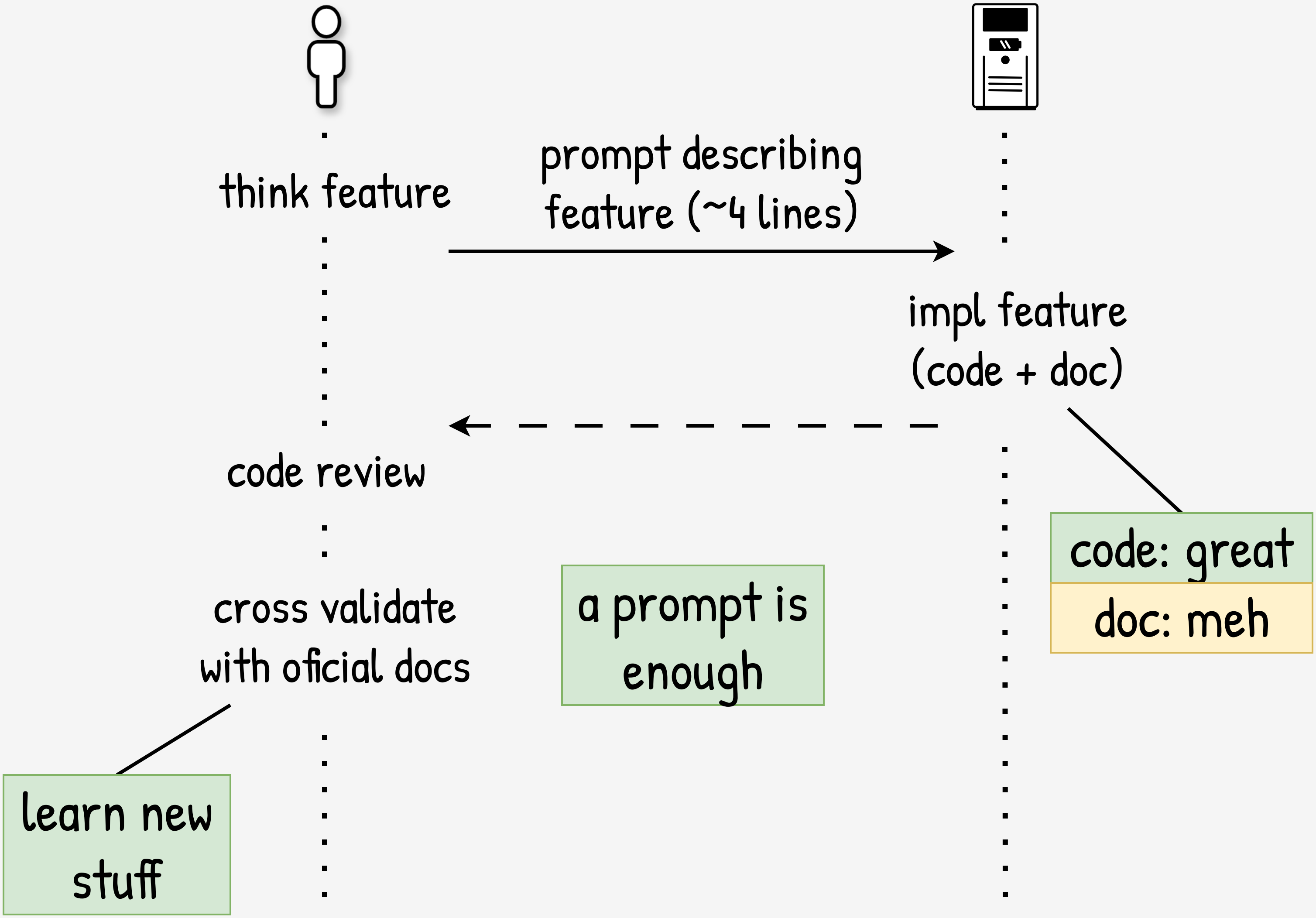



think feature
prompt describing
feature (~4 lines)
impl feature
(code + doc)

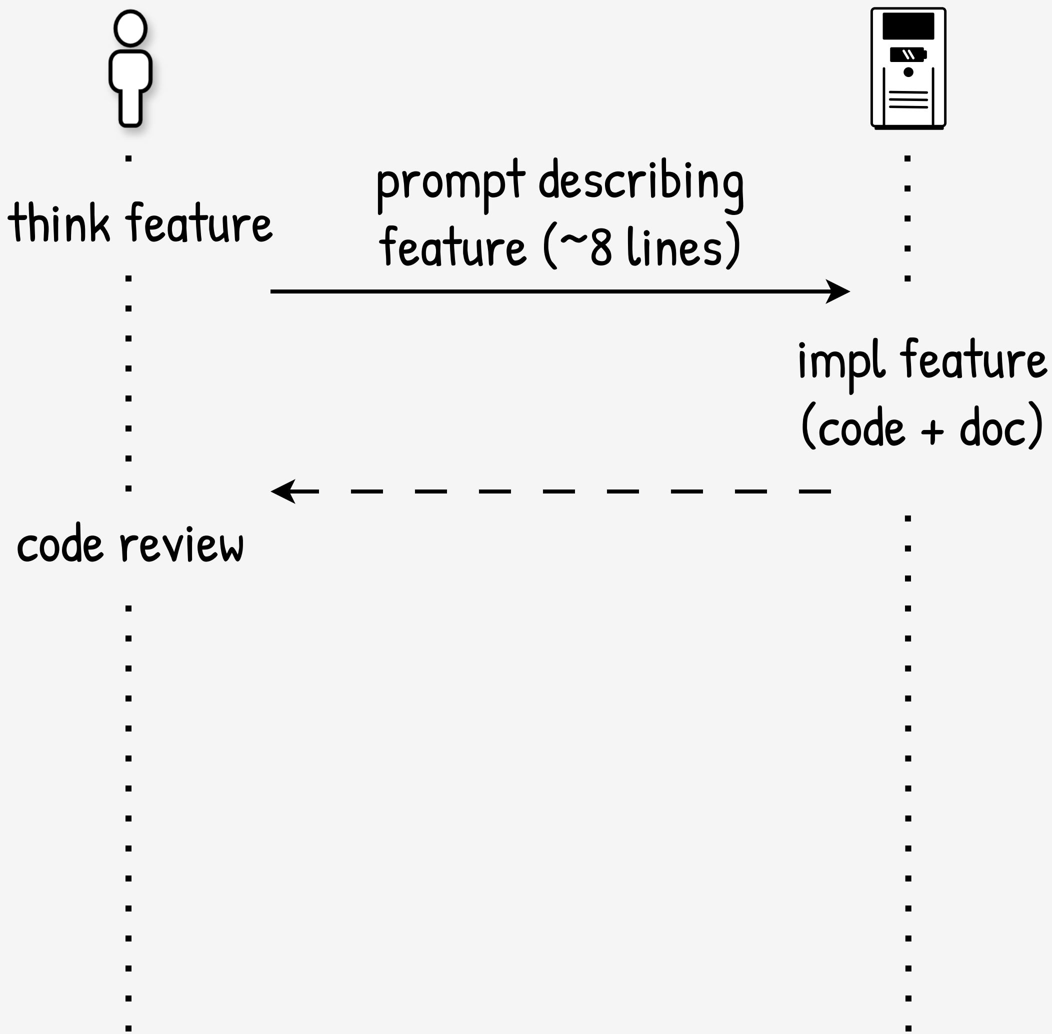

think feature
prompt describing
feature (~4 lines)
impl feature
(code + doc)
code review
cross validate
with oficial docs

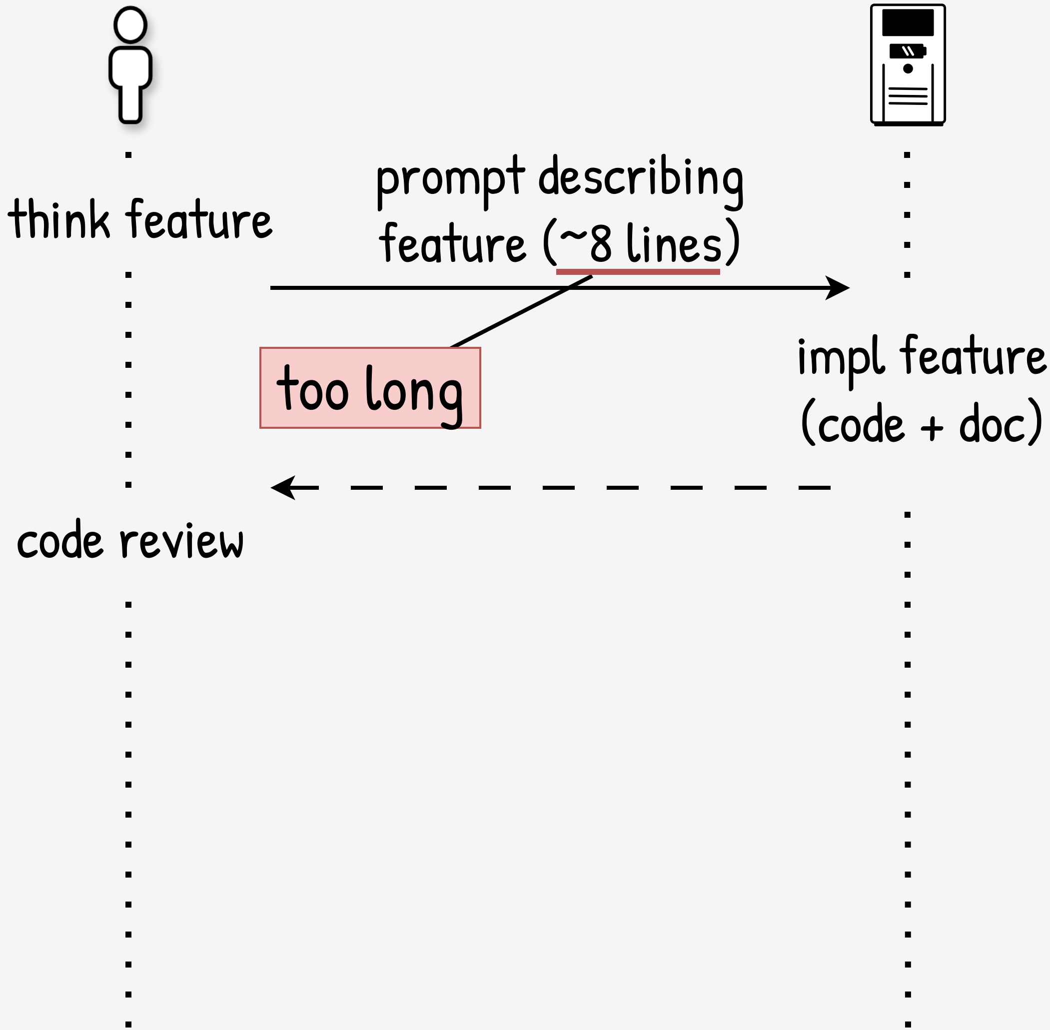

think feature
prompt describing feature (~4 lines)
impl feature (code + doc)
code review
cross validate with oficial docs
code: great
doc: meh

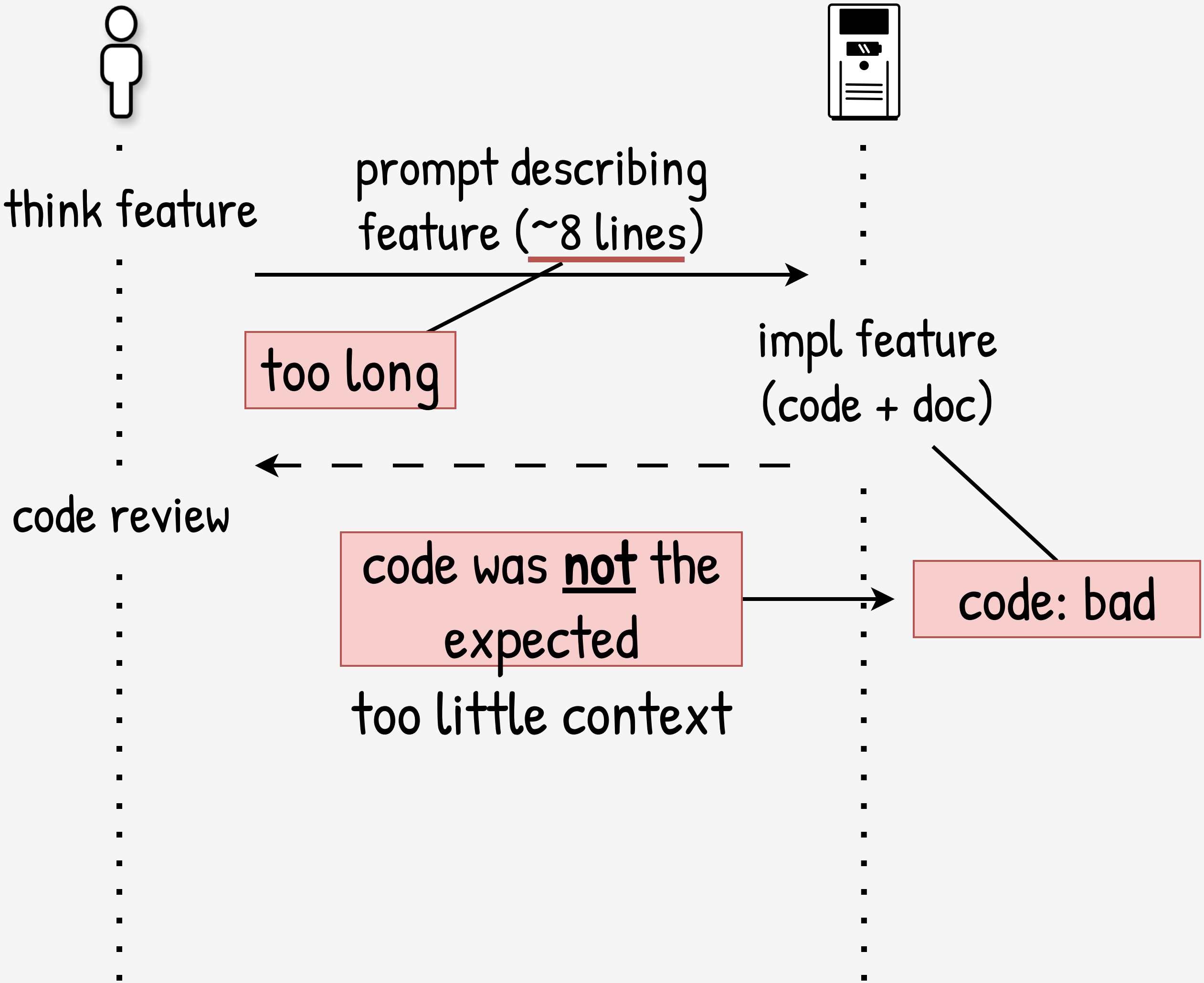

think feature
prompt describing feature (~4 lines)
impl feature (code + doc)
code review
code: great
doc: meh
cross validate with oficial docs
learn new stuff

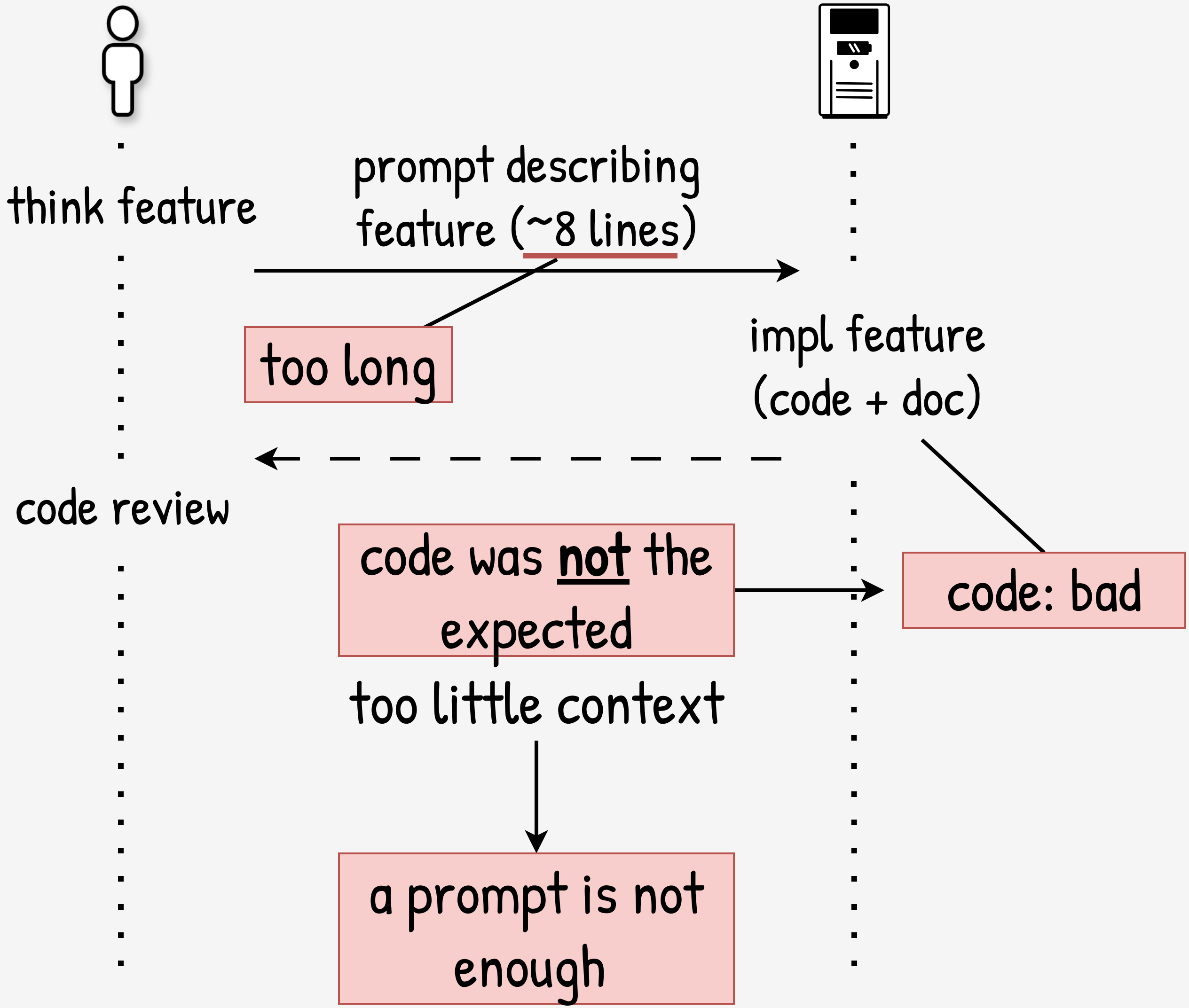


think feature
prompt describing feature (~4 lines)
impl feature (code + doc)
code: great
doc: meh
code review
cross validate with oficial docs
a prompt is enough
learn new stuff

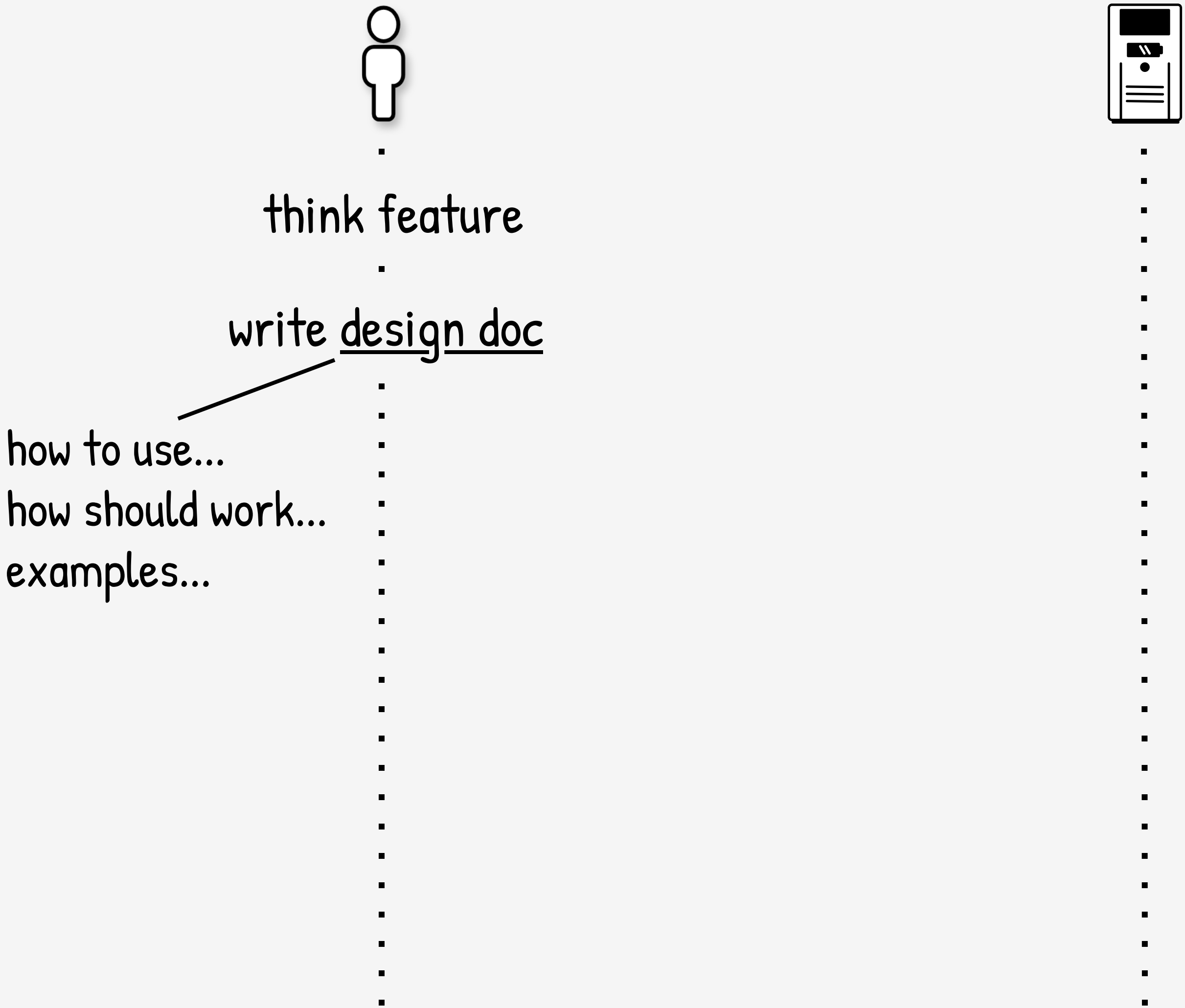

think feature
prompt describing
feature (~8 lines)
impl feature
(code + doc)
code review

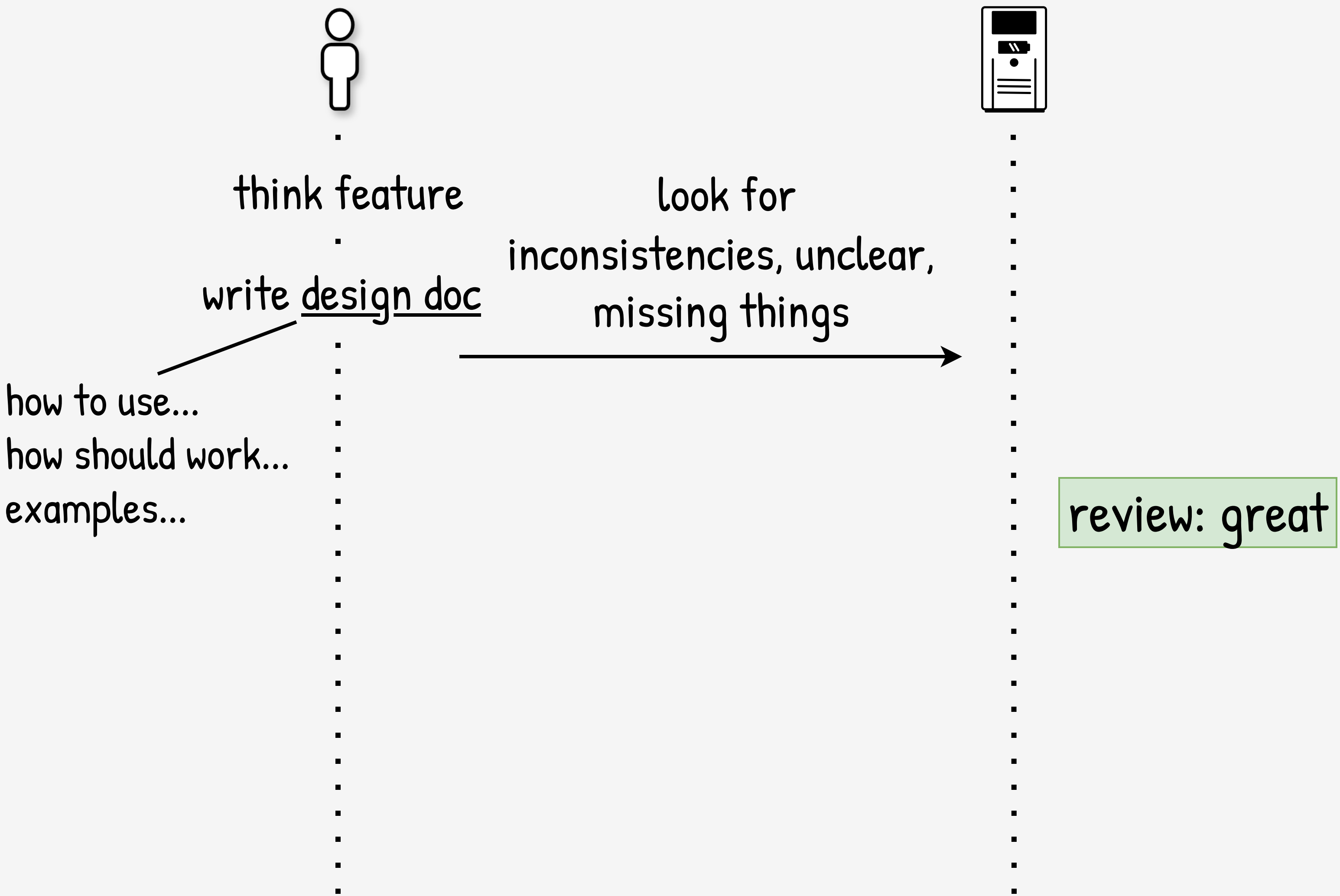

think feature
prompt describing
feature (~8 lines)
too long
impl feature
(code + doc)
code review

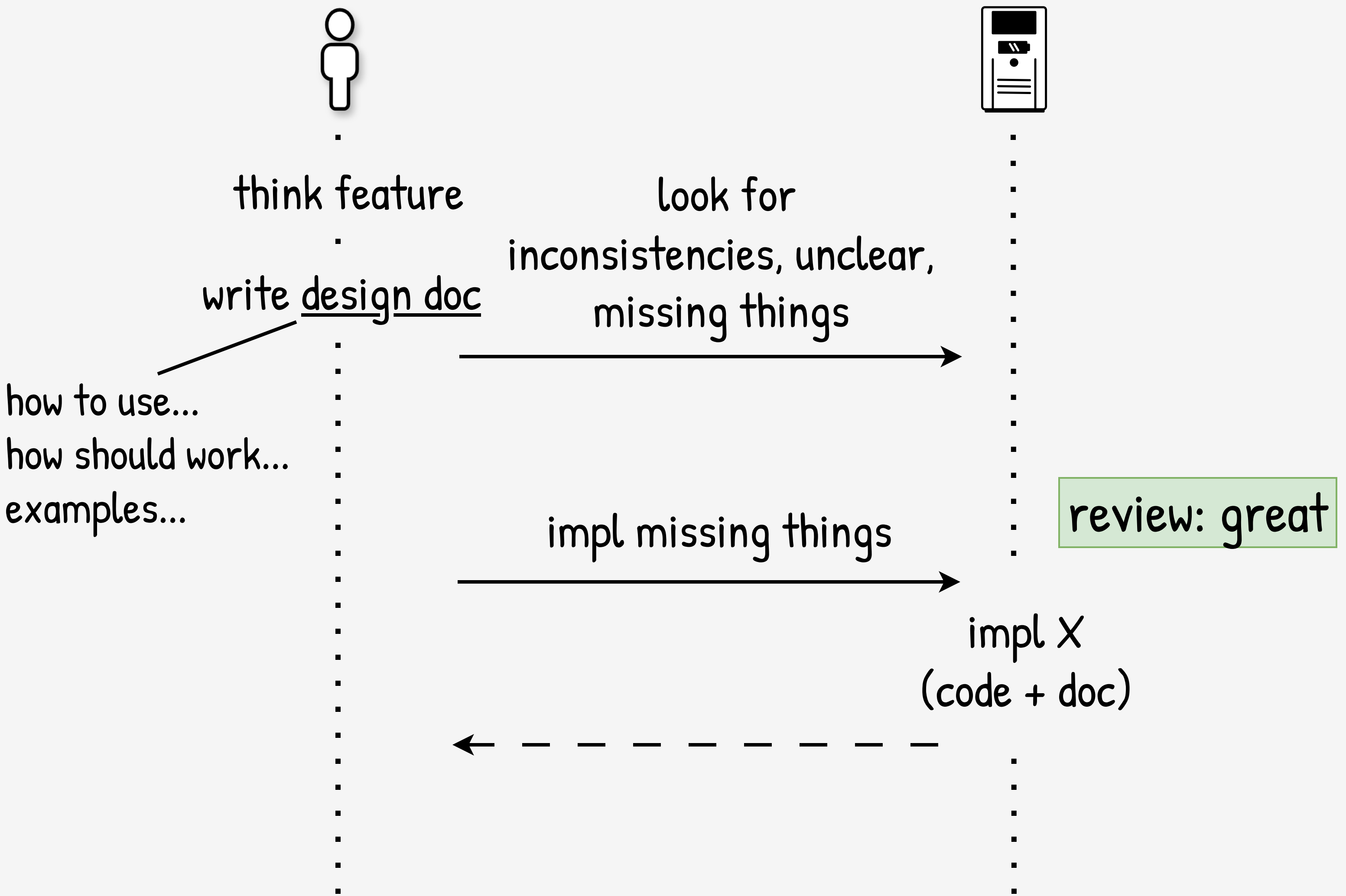

think feature
prompt describing feature (~8 lines)
too long
impl feature (code + doc)
code review
code was **not** the expected
code: bad
too little context

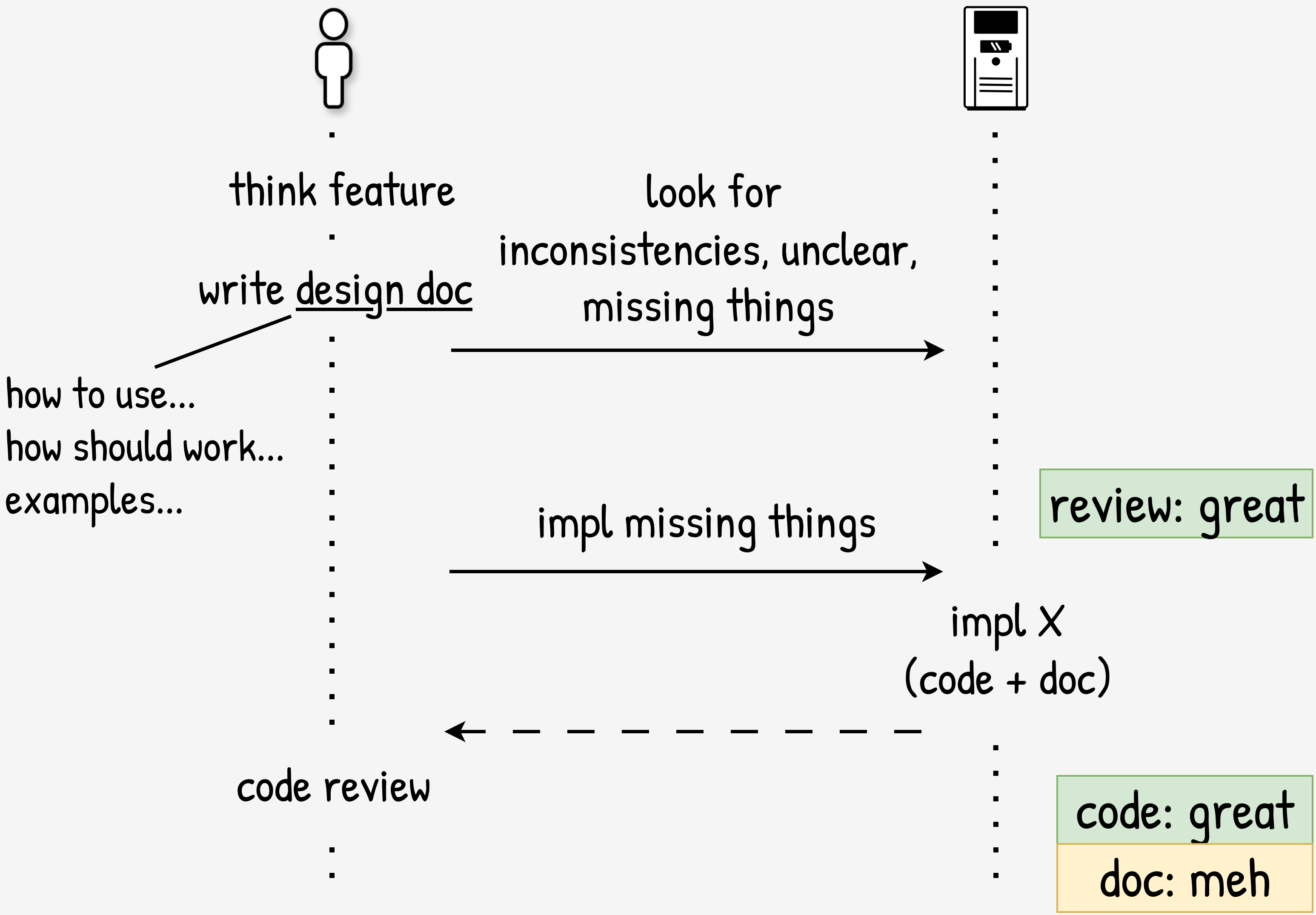


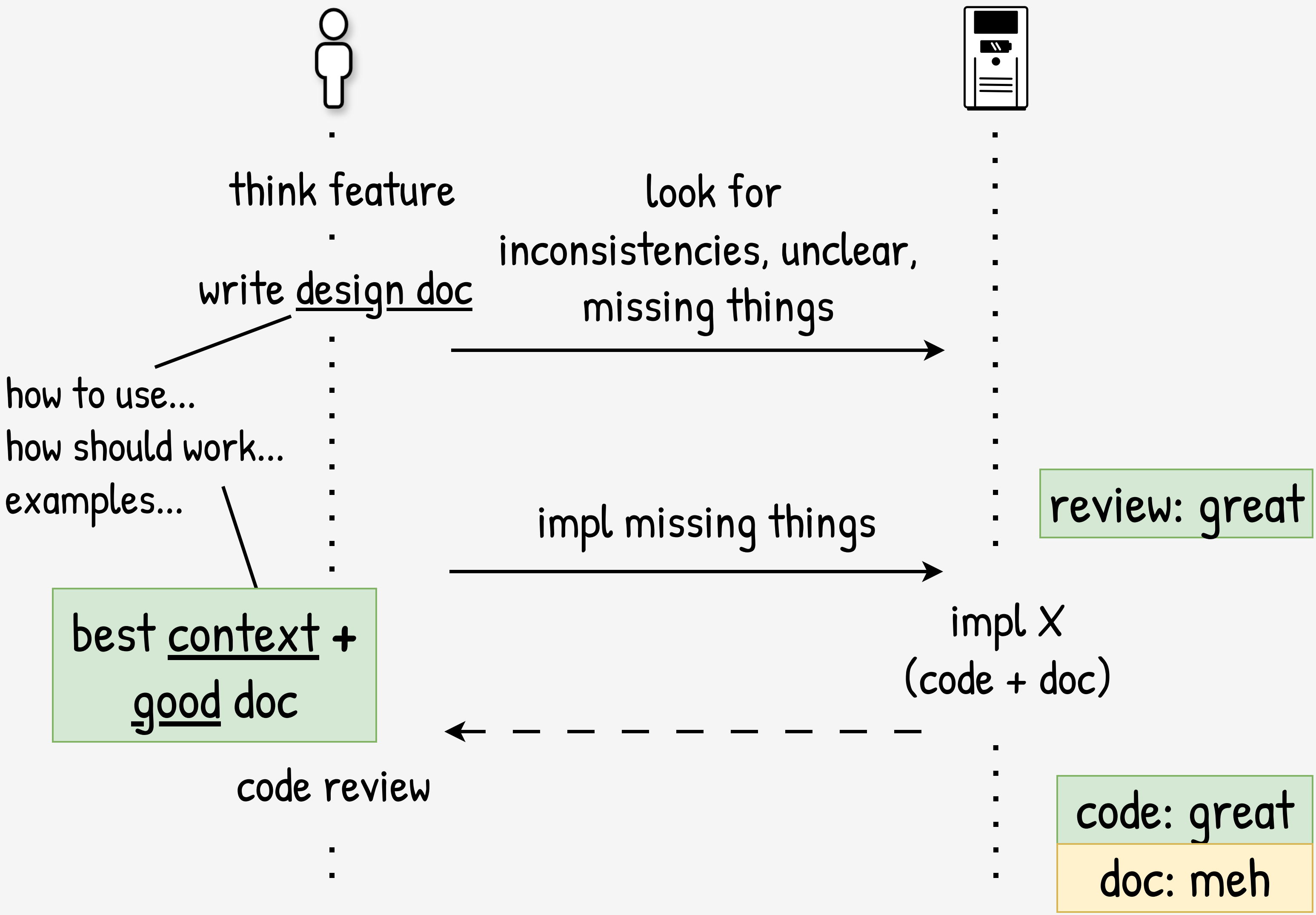
think feature
prompt describing feature (~8 lines)
too long
impl feature (code + doc)
code review
code was **not** the expected
code: bad
too little context
a prompt is not enough


think feature
write design doc
how to use...
how should work...
examples...

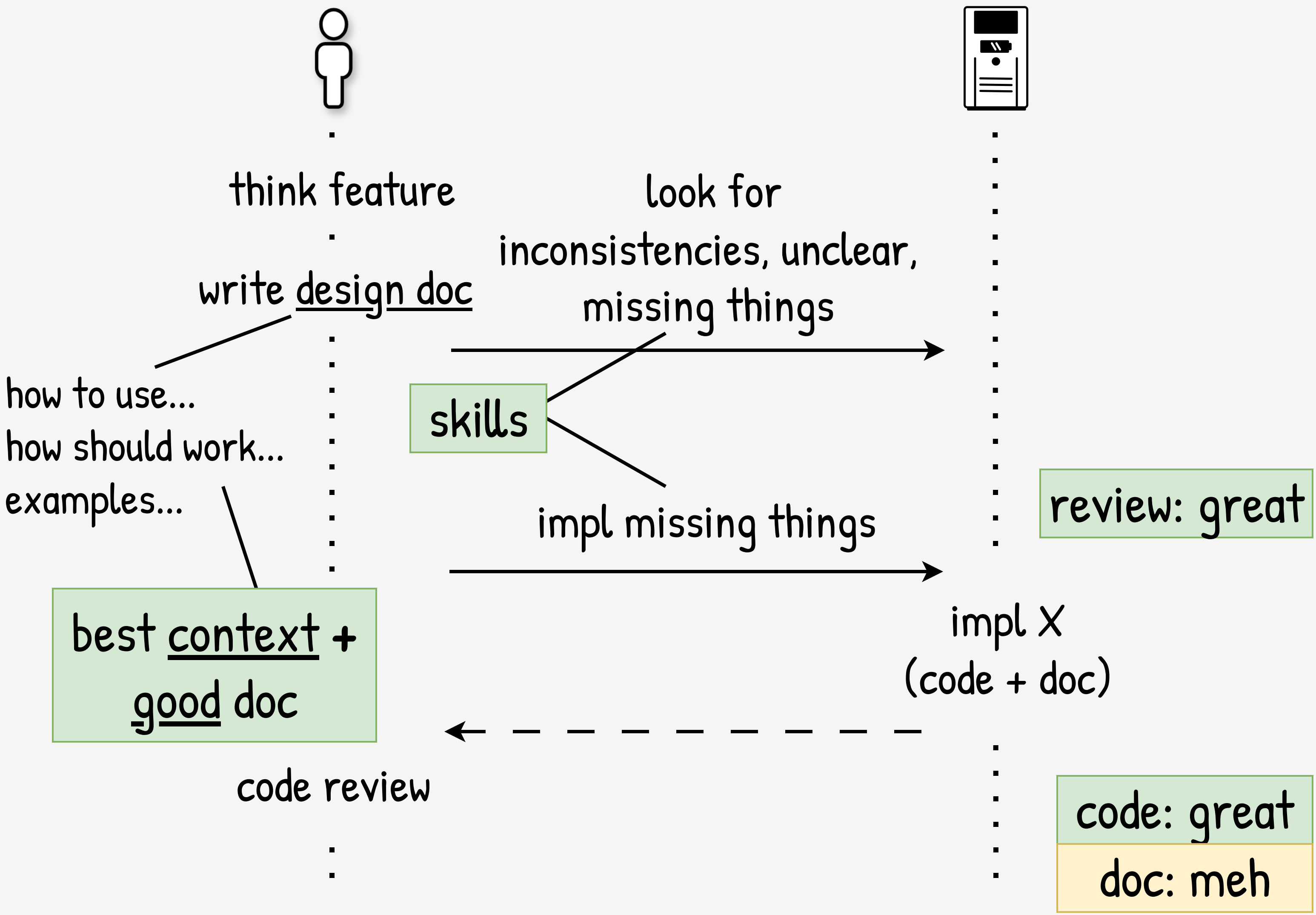

think feature
write design doc
how to use...
how should work...
examples...
look for
inconsistencies, unclear,
missing things
review: great

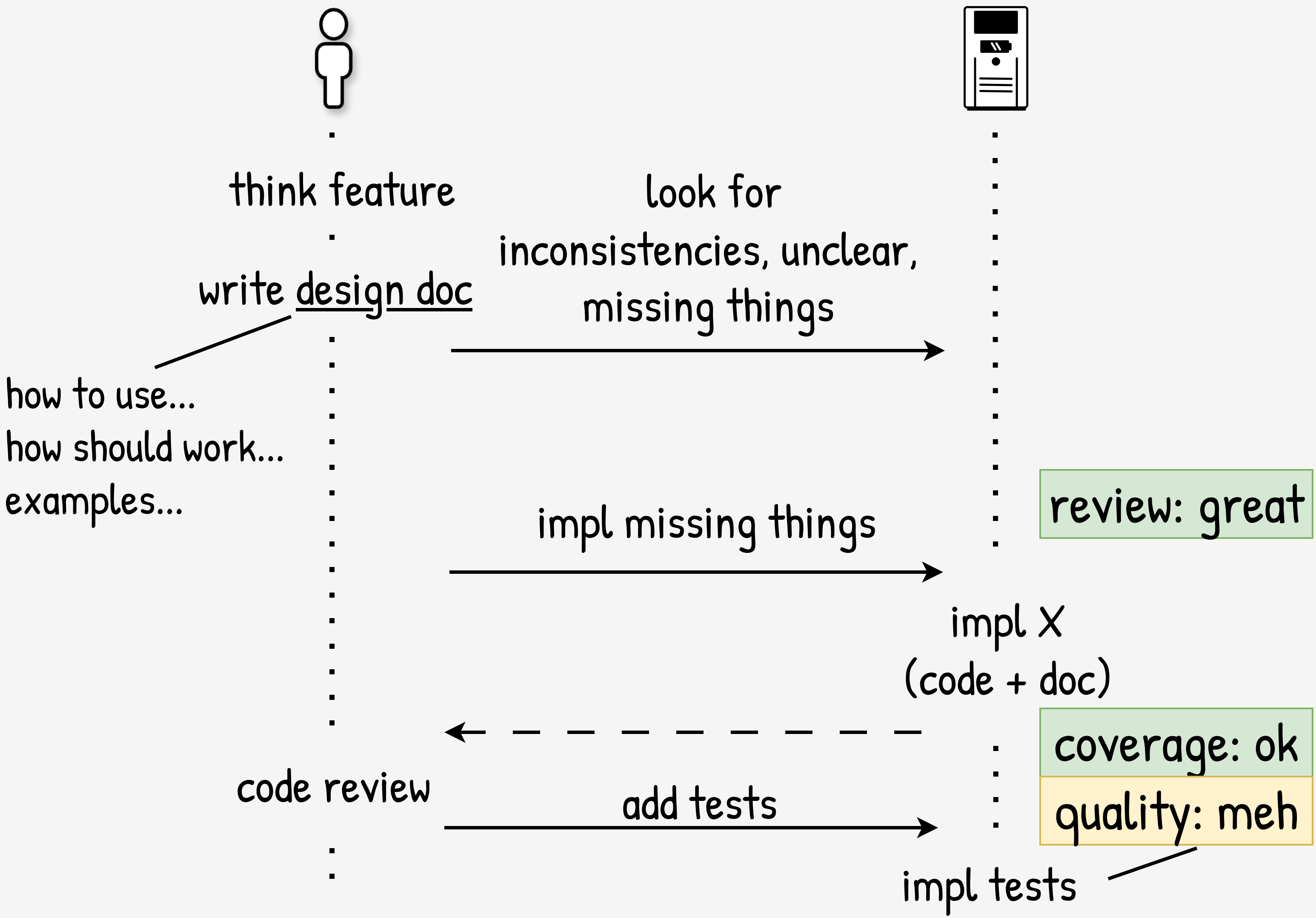

think feature
write design doc
how to use...
how should work...
examples...
look for
inconsistencies, unclear,
missing things
impl missing things
review: great
impl X
(code + doc)


think feature
write design doc
how to use...
how should work...
examples...
look for
inconsistencies, unclear,
missing things
review: great
impl missing things
impl X
(code + doc)
code review
code: great
doc: meh



think feature
write design doc
how to use...
how should work...
examples...
best context + good doc
look for inconsistencies, unclear, missing things
impl missing things
review: great
impl X (code + doc)
code review
code: great
doc: meh


think feature
look for
inconsistencies, unclear,
missing things
write design doc
how to use...
how should work...
examples...
skills
review: great
impl missing things
best context +
good doc
impl X
(code + doc)
code review
code: great
doc: meh


think feature
look for
inconsistencies, unclear,
missing things
write design doc
how to use...
how should work...
examples...
impl missing things
review: great
impl X
(code + doc)
coverage: ok
quality: meh
code review
add tests
impl tests

feature <u>refinement</u>

- non-trivial changes
- a lot of refactors
- integration with other features

pass of time, refinements and iterations

review doc

code

doc code

test coverage

test quality

<u>my</u> understanding

pass of time, refinements and iterations →

review doc	great	*awesome*
code		
doc code		
test coverage		
test quality		
my understanding		

pass of time, refinements and iterations →

review doc	great	*awesome*
code	great	weird
doc code	meh	useless
test coverage		
test quality		
<u>my</u> understanding		

pass of time, refinements and iterations →

review doc	great	*awesome*	
code	great	weird	
doc code	meh	useless	
test coverage	ok	maybe ok?	>5k lines
test quality	meh	unintelligible	a lot of mocks
<u>my</u> understanding			

pass of time, refinements and iterations

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my understanding	ok	I have no idea	

pass of time, refinements and iterations →

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my understanding	ok	I have no idea	

I lost the capability to do code review and abstraction → I was **out** of the loop

pass of time, refinements and iterations →

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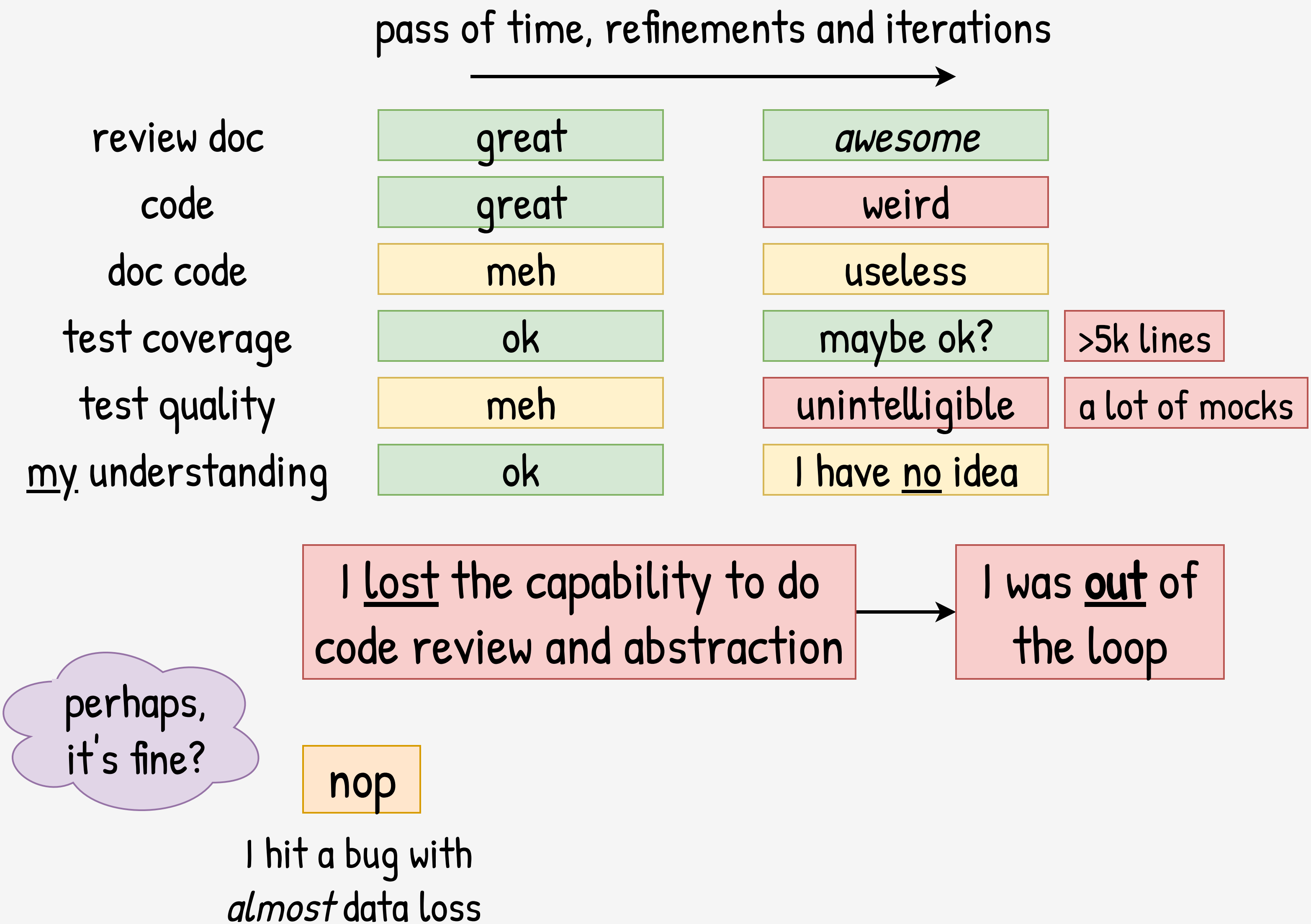
<u>my</u> understanding	ok	I have <u>no</u> idea	

I <u>lost</u> the capability to do code review and abstraction → I was **<u>out</u>** of the loop

perhaps, it's fine?


pass of time, refinements and iterations
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test coverage
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test quality
meh
unintelligible
a lot of mocks
my understanding
ok
I have no idea
I lost the capability to do code review and abstraction
I was **out** of the loop
perhaps, it's fine?
nop
I hit a bug with *almost* data loss

pass of time, refinements and iterations →

review doc	great	*awesome*	
code	great	weird	
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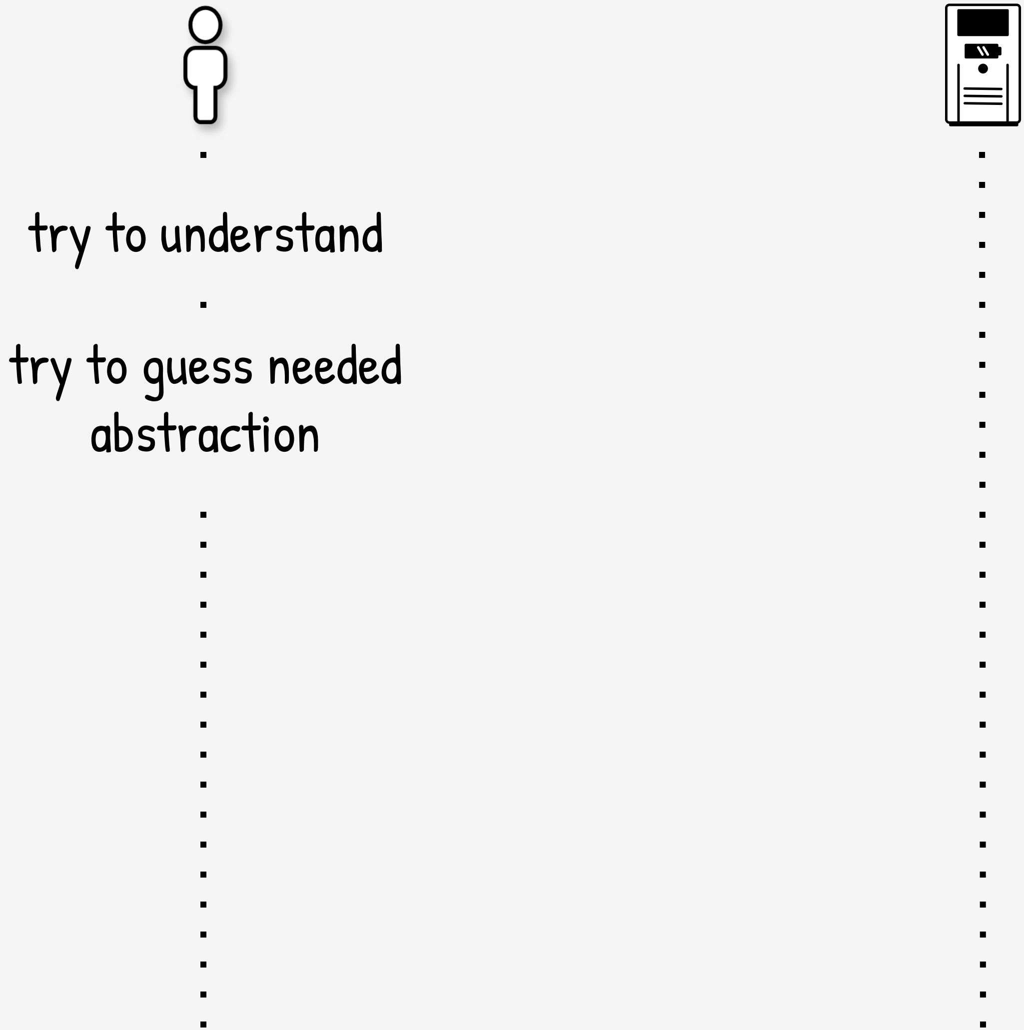
my understanding	ok	I have no idea	

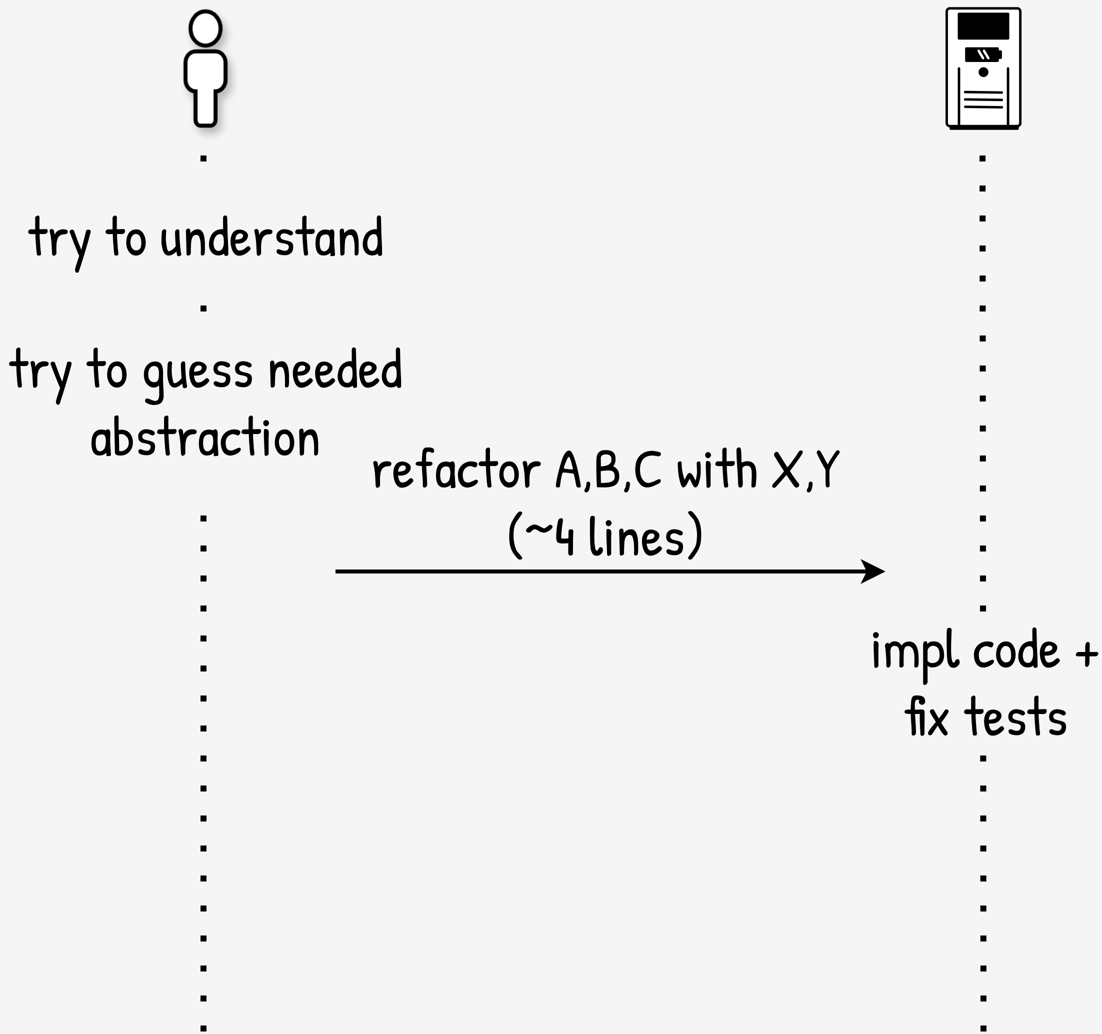
I lost the capability to do code review and abstraction → I was **out** of the loop

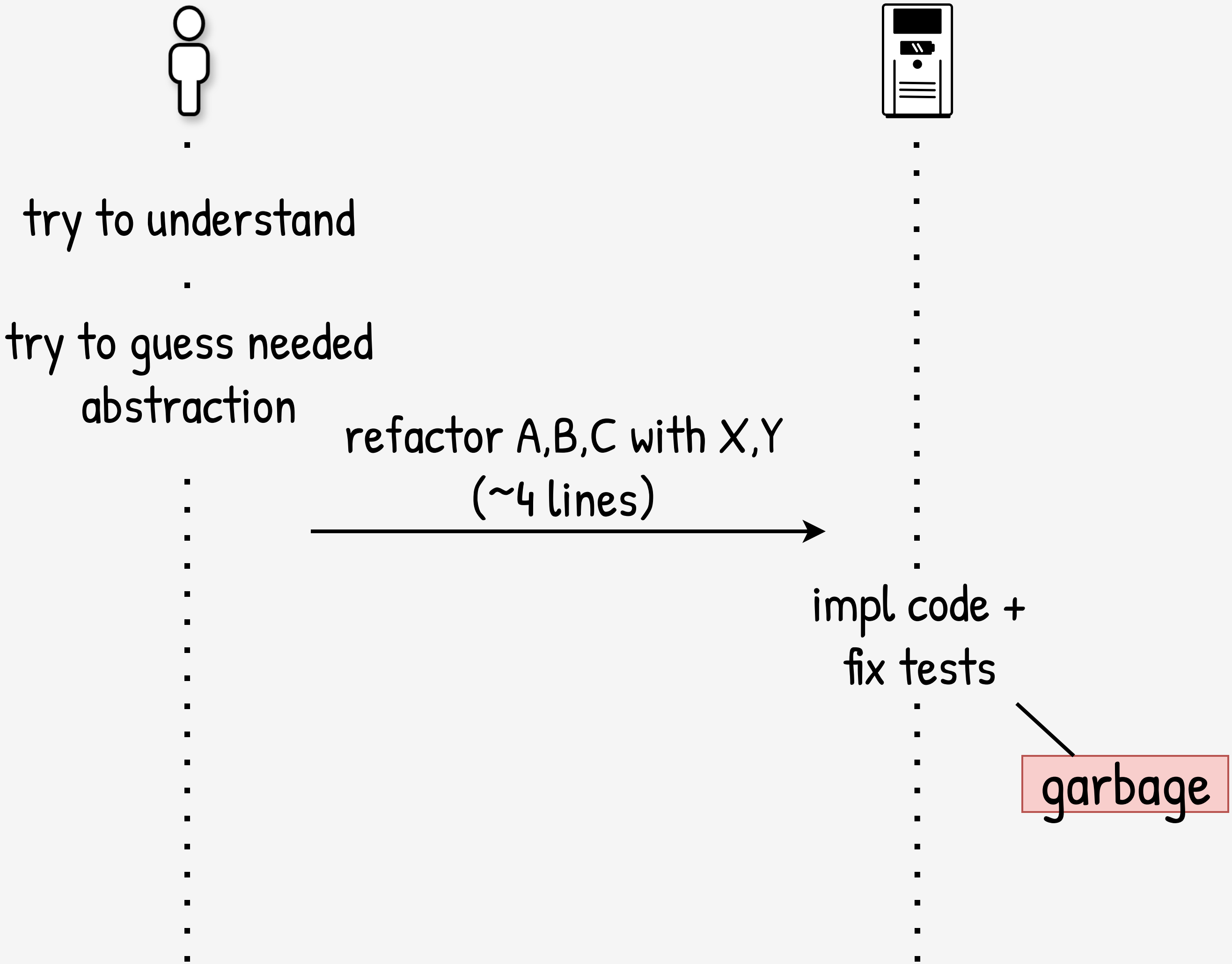
perhaps, it's fine?

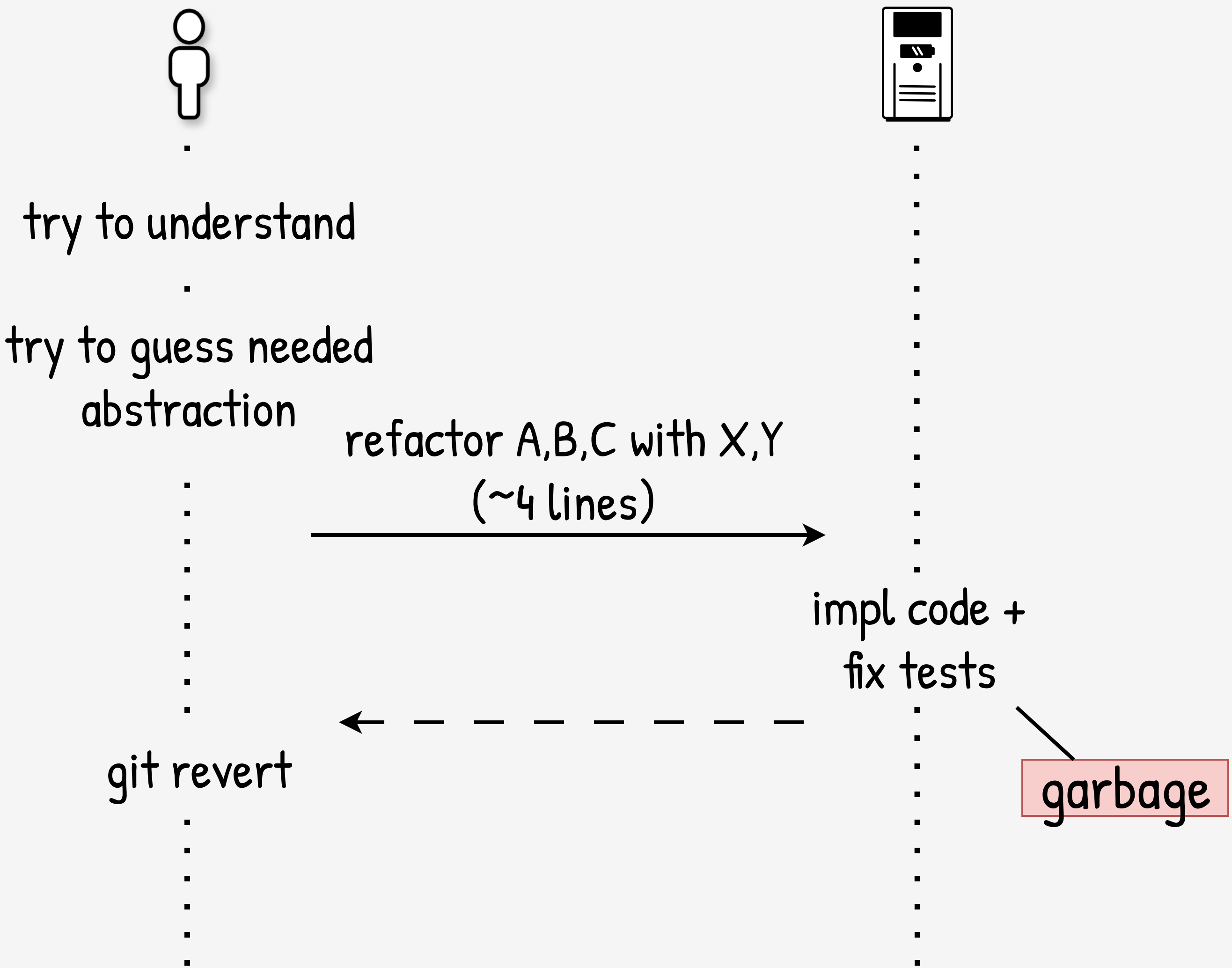
nop → design docs are incomplete, they are **not** a spec → the AI fills the gaps without asking

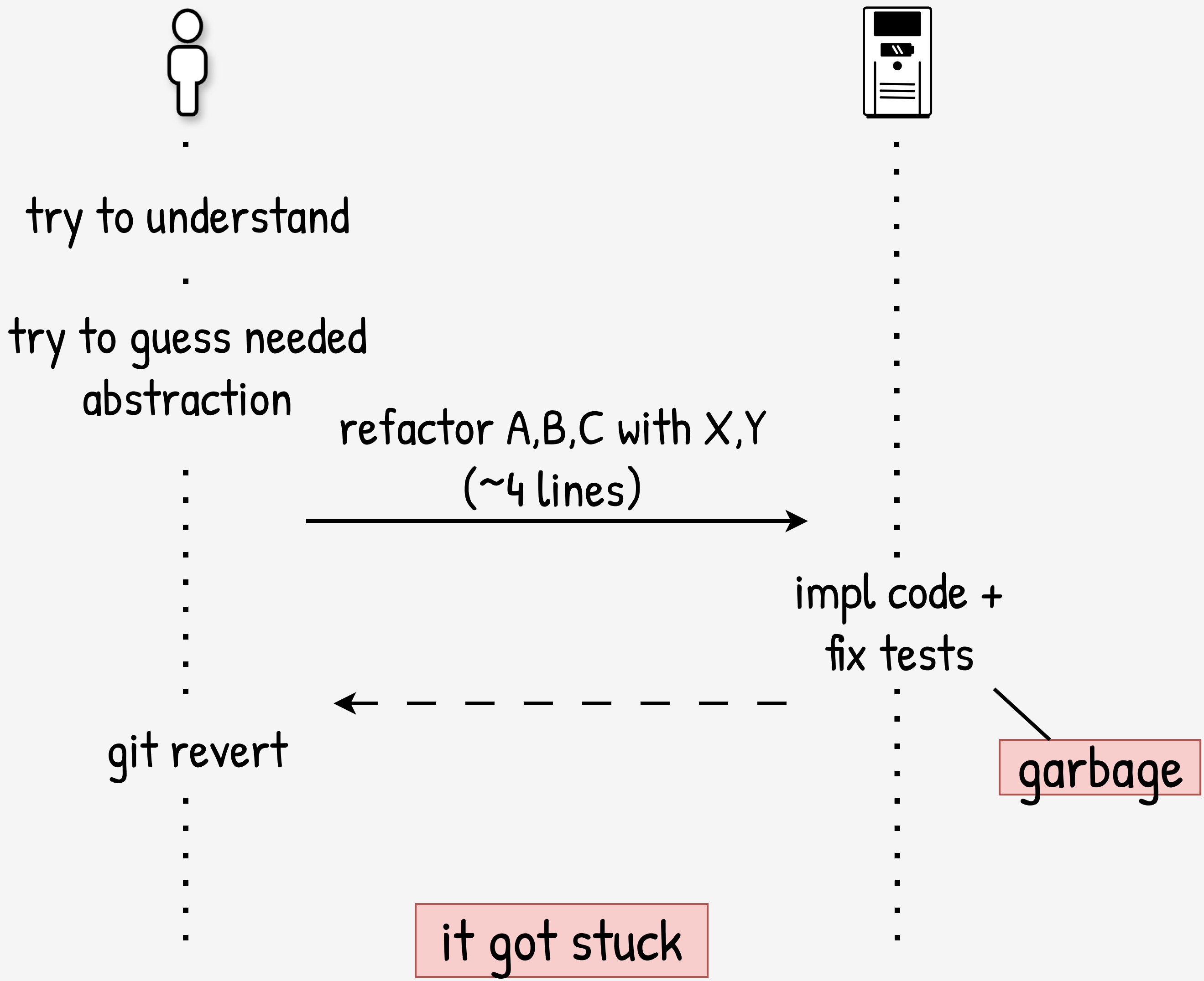
I hit a bug with *almost* data loss

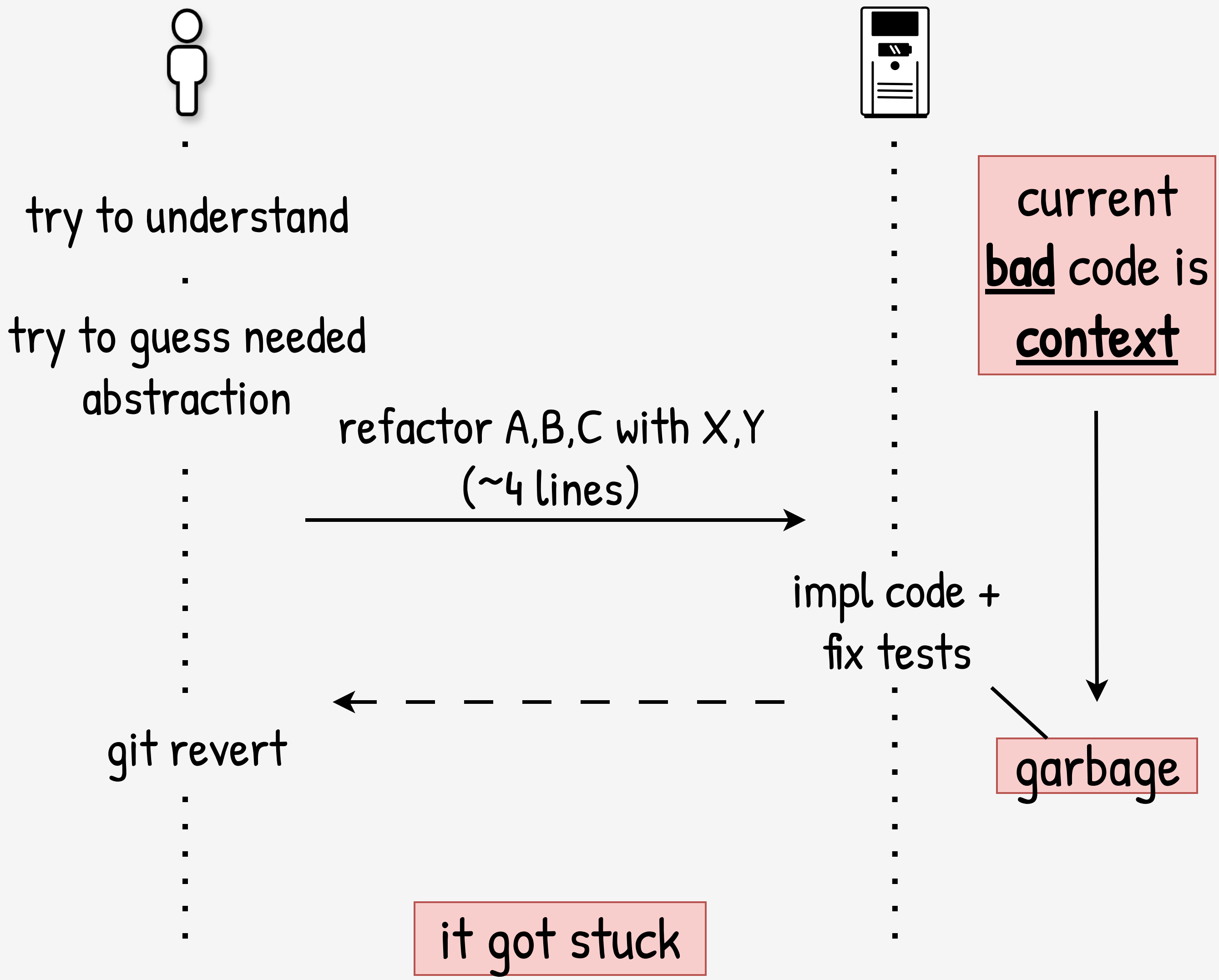

try to understand
try to guess needed abstraction

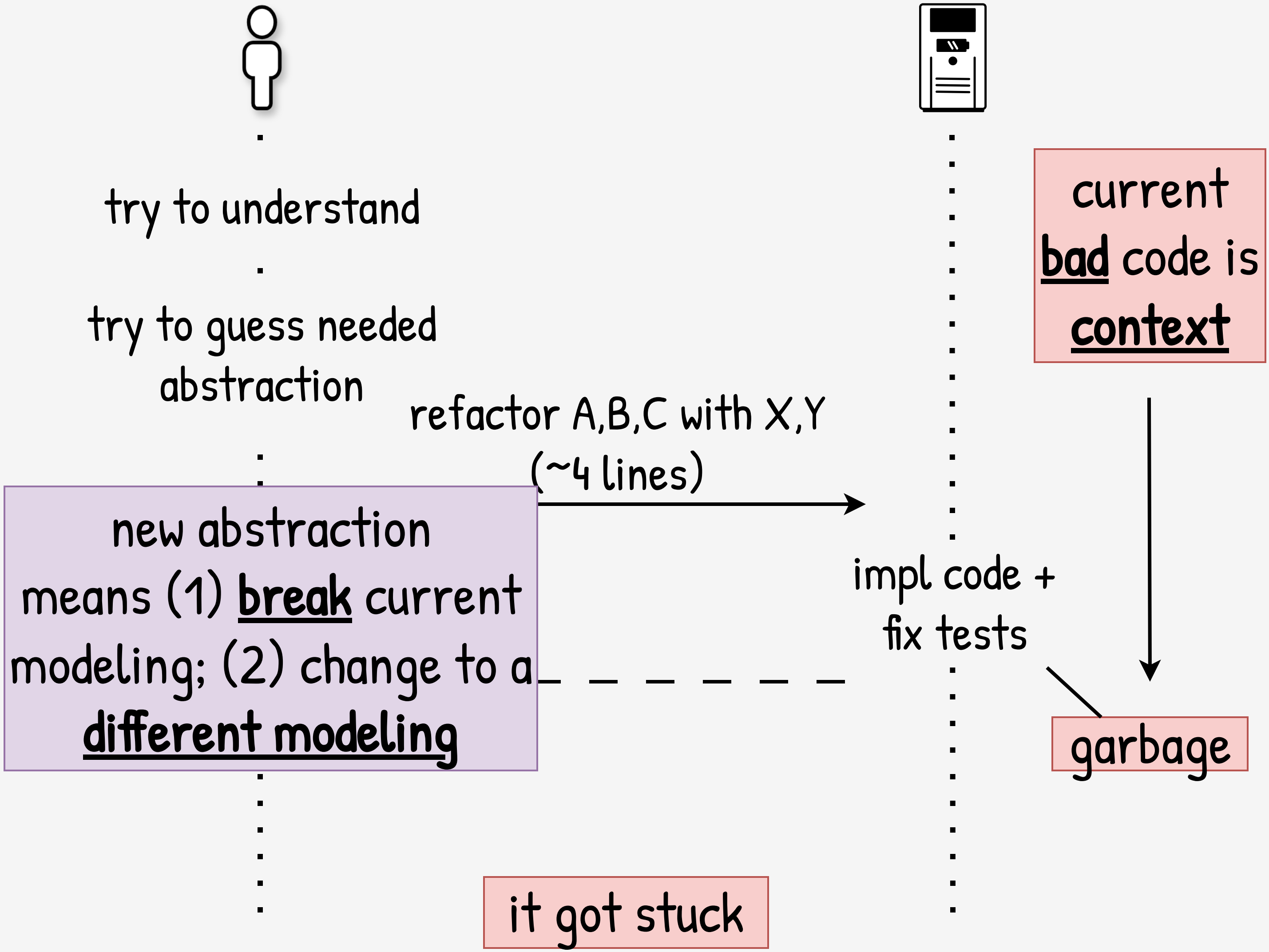

try to understand
try to guess needed abstraction
refactor A,B,C with X,Y (~4 lines)
impl code + fix tests


try to understand
try to guess needed abstraction
refactor A,B,C with X,Y (~4 lines)
impl code + fix tests
garbage

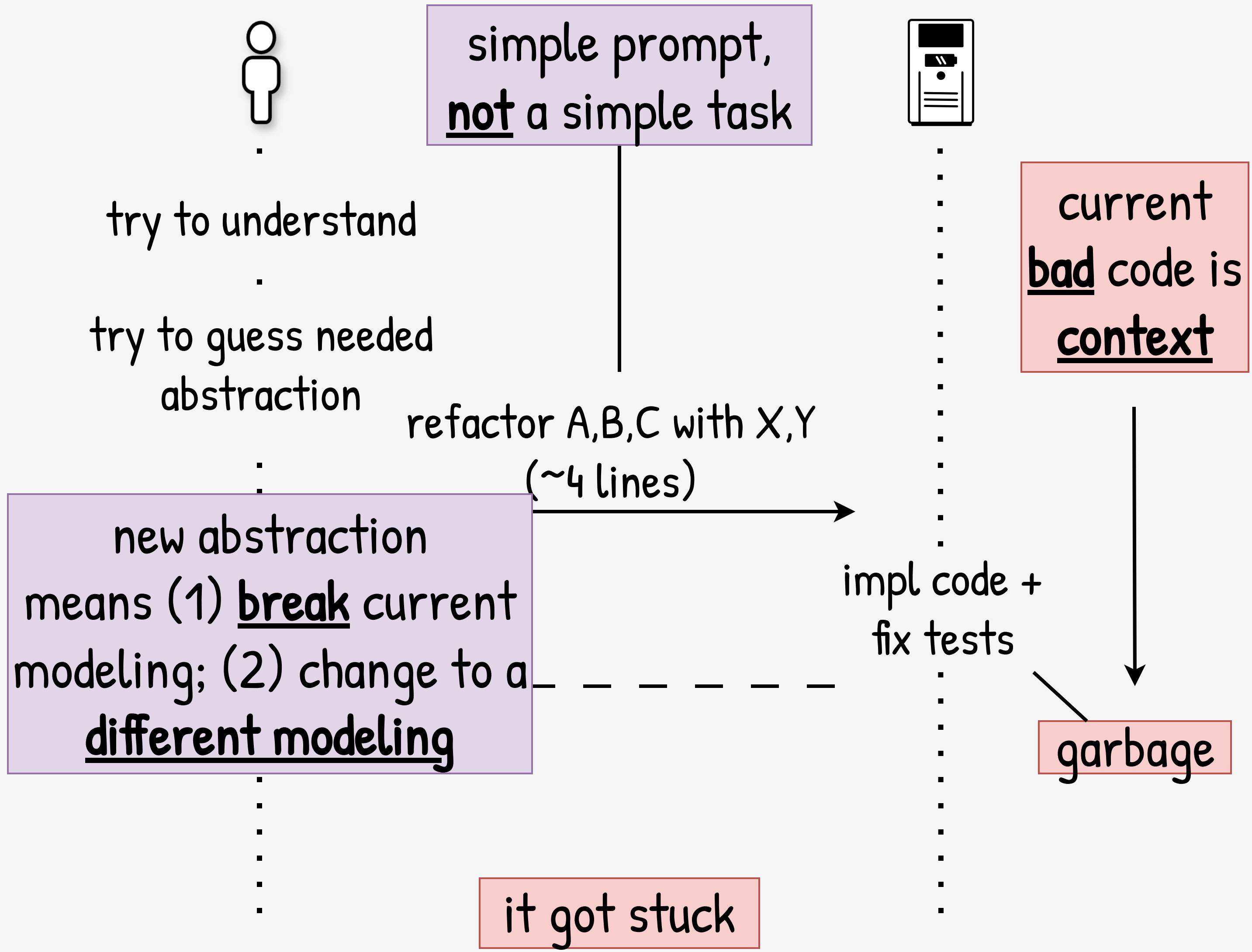

try to understand
try to guess needed abstraction
refactor A,B,C with X,Y (~4 lines)
impl code + fix tests
garbage
git revert


try to understand
try to guess needed abstraction
refactor A,B,C with X,Y (~4 lines)
impl code + fix tests
git revert
garbage
it got stuck

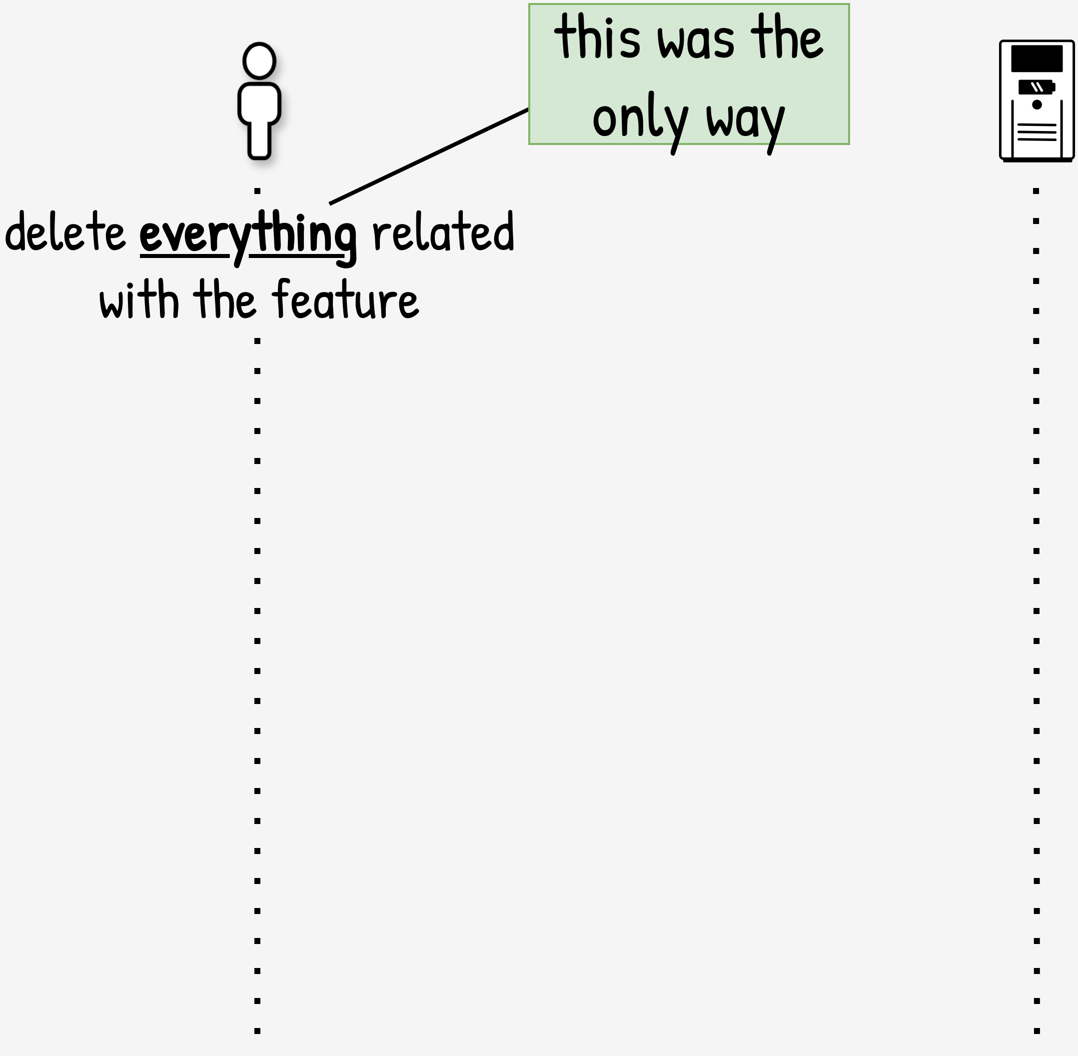

try to understand
try to guess needed abstraction
refactor A,B,C with X,Y (~4 lines)
impl code + fix tests
git revert
current **bad** code is **context**
garbage
it got stuck

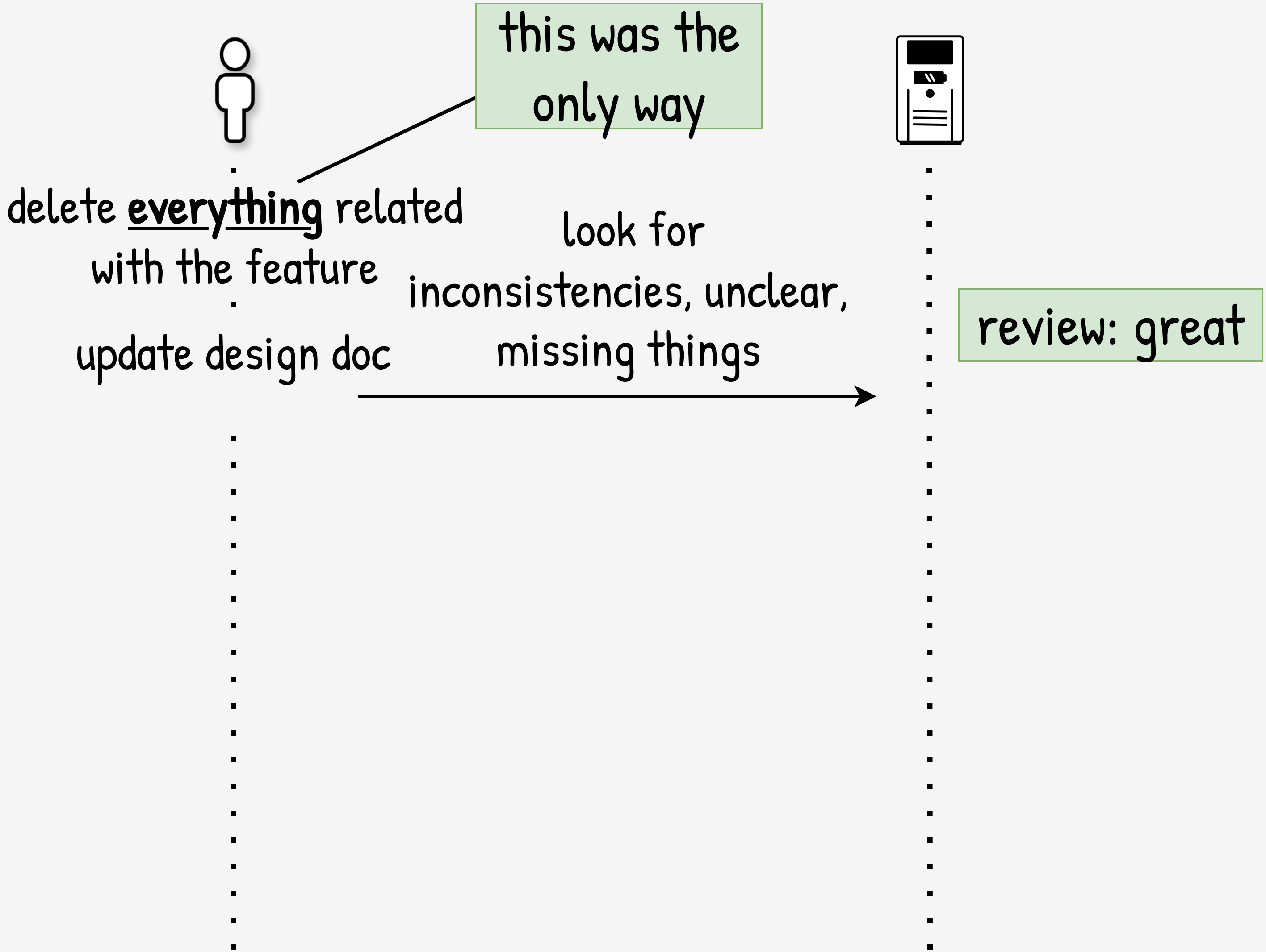


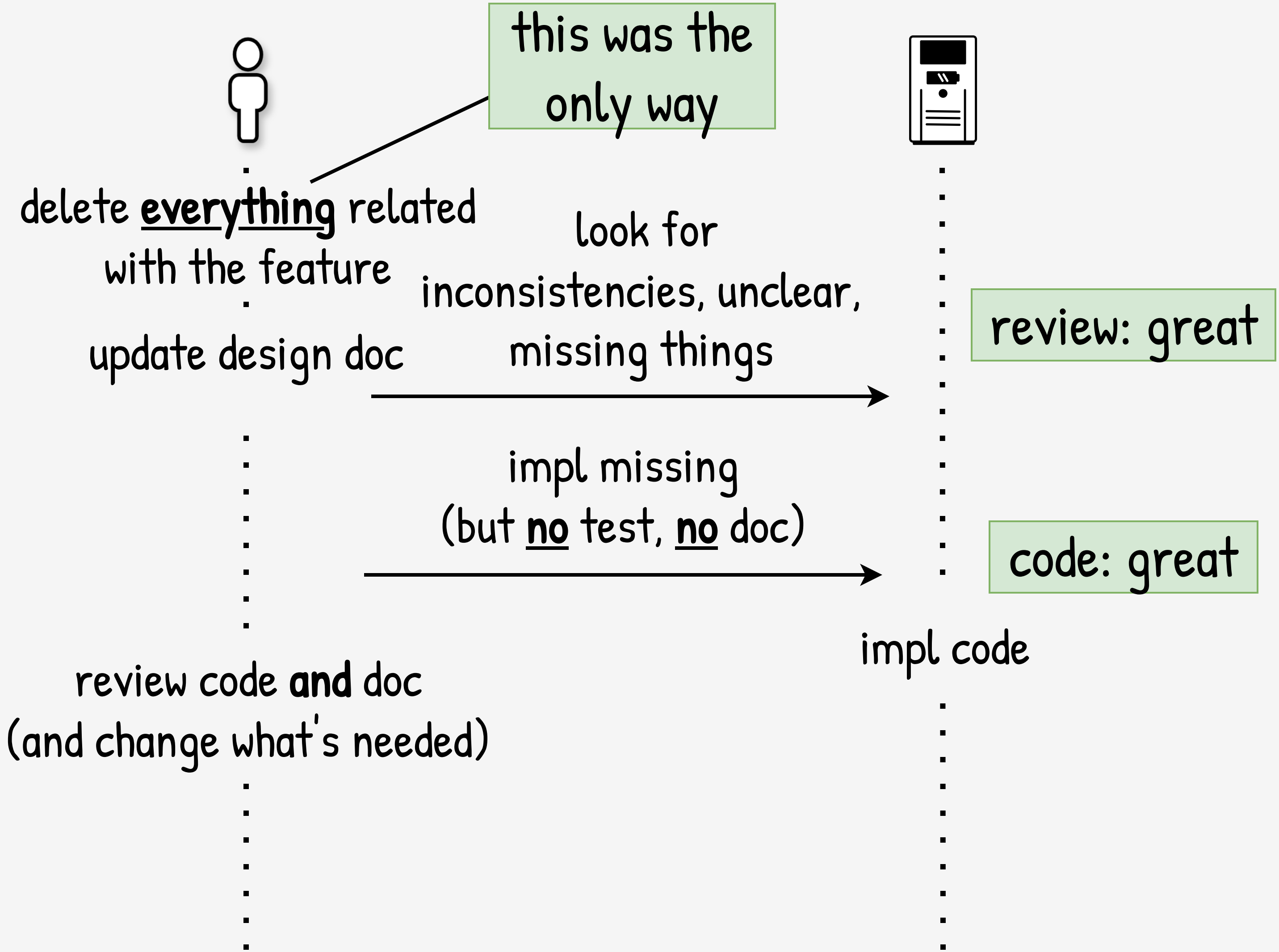
try to understand
try to guess needed abstraction
refactor A,B,C with X,Y (~4 lines)
new abstraction means (1) break current modeling; (2) change to a different modeling
impl code + fix tests
current bad code is context
garbage
it got stuck

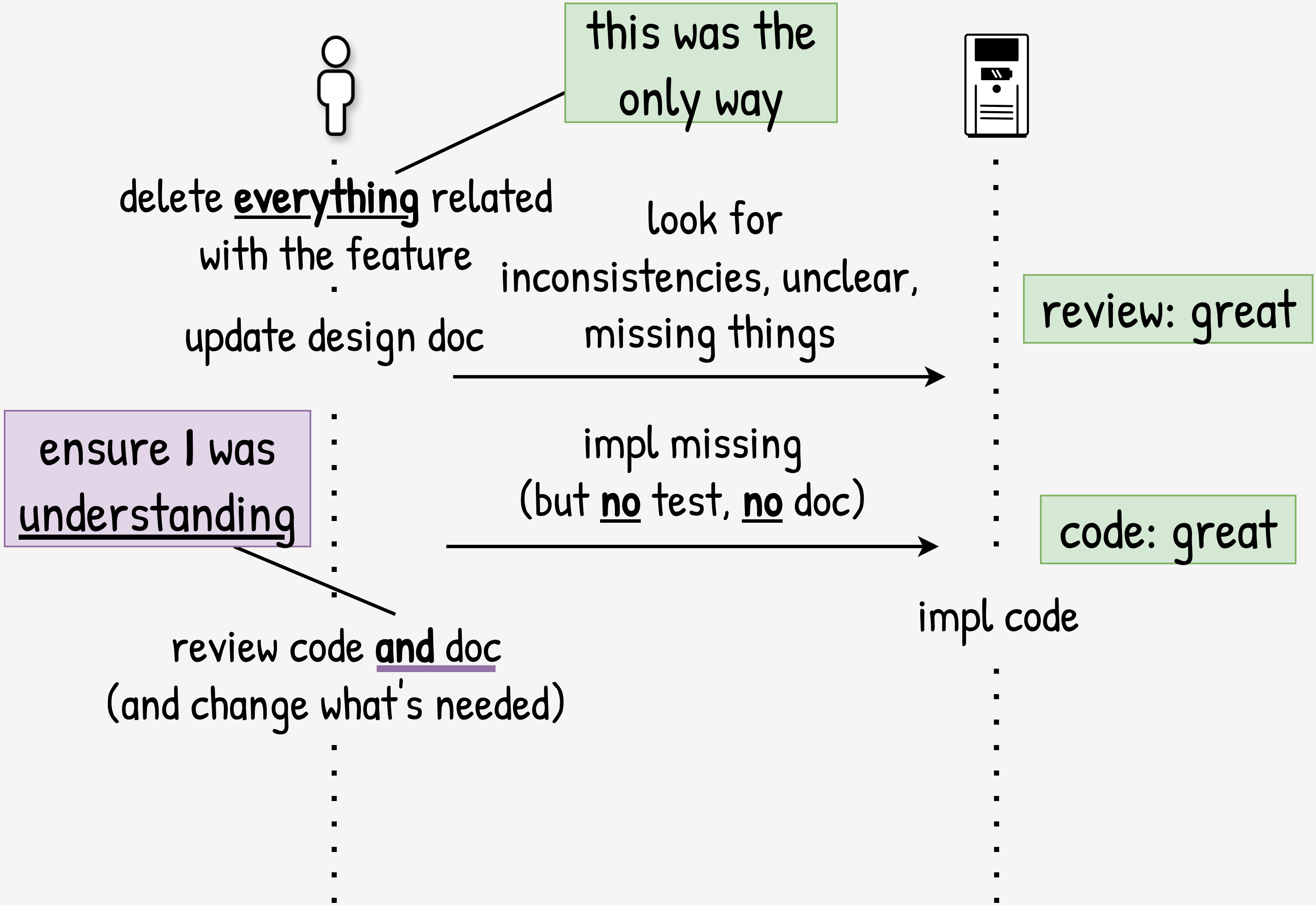


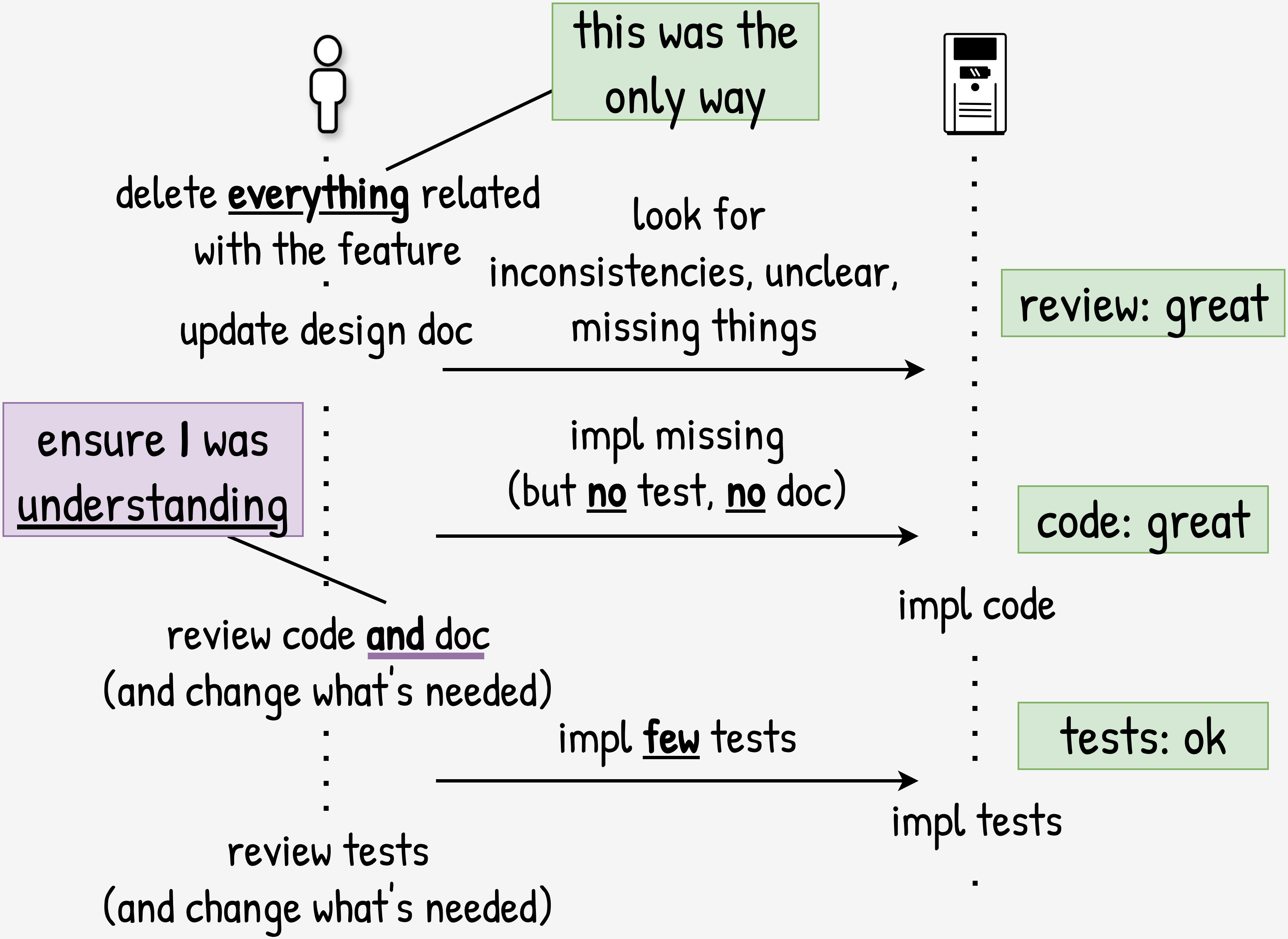
simple prompt,
not a simple task
try to understand
try to guess needed abstraction
refactor A,B,C with X,Y (~4 lines)
current **bad** code is **context**
new abstraction means (1) **break** current modeling; (2) change to a **different modeling**
impl code + fix tests
garbage
it got stuck

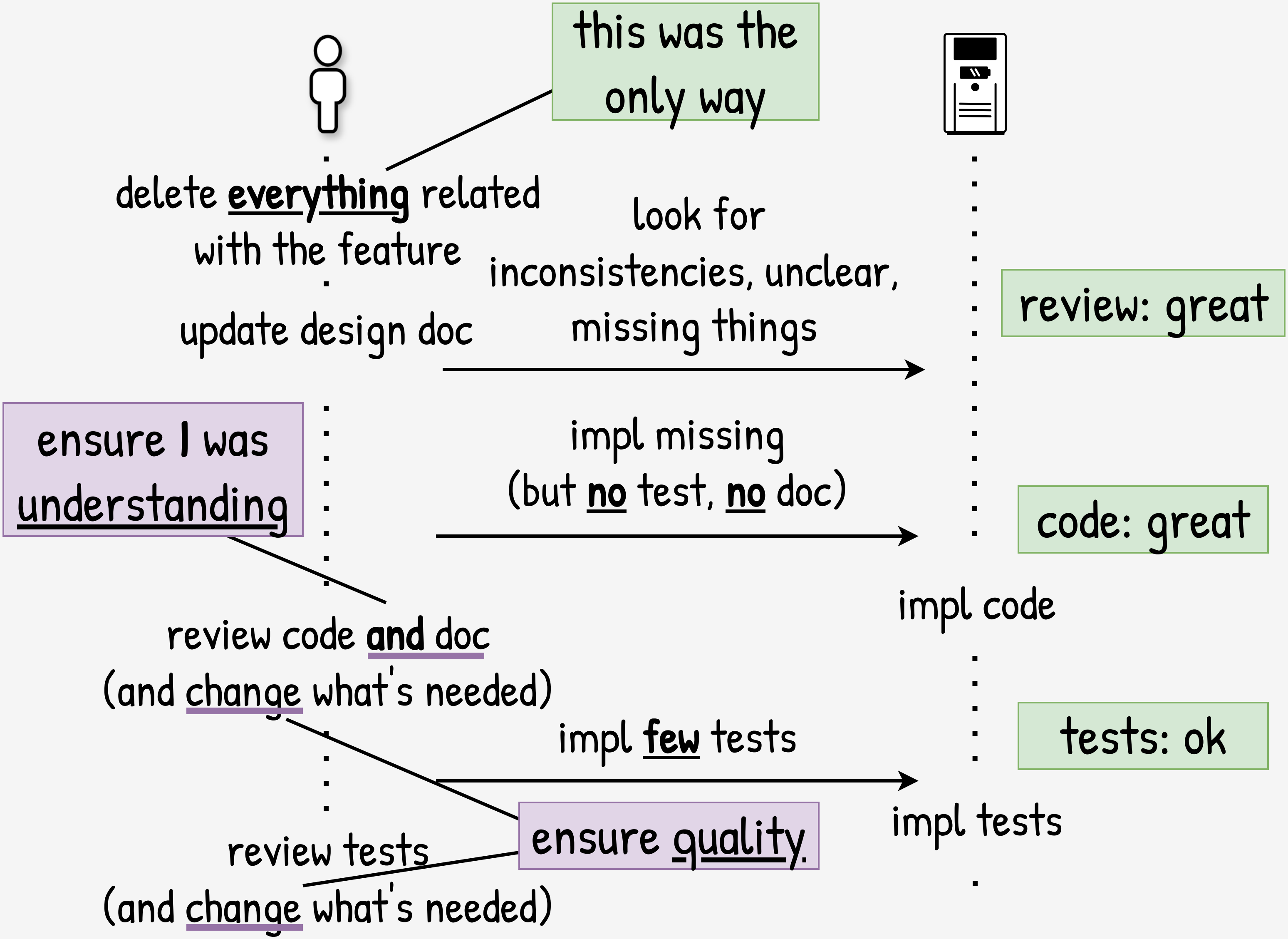

this was the only way
delete **everything** related with the feature


this was the only way
delete **everything** related with the feature
update design doc
look for inconsistencies, unclear, missing things
review: great


this was the only way
delete **everything** related with the feature
update design doc
look for inconsistencies, unclear, missing things
review: great
impl missing (but **no** test, **no** doc)
code: great
impl code
review code **and** doc (and change what's needed)

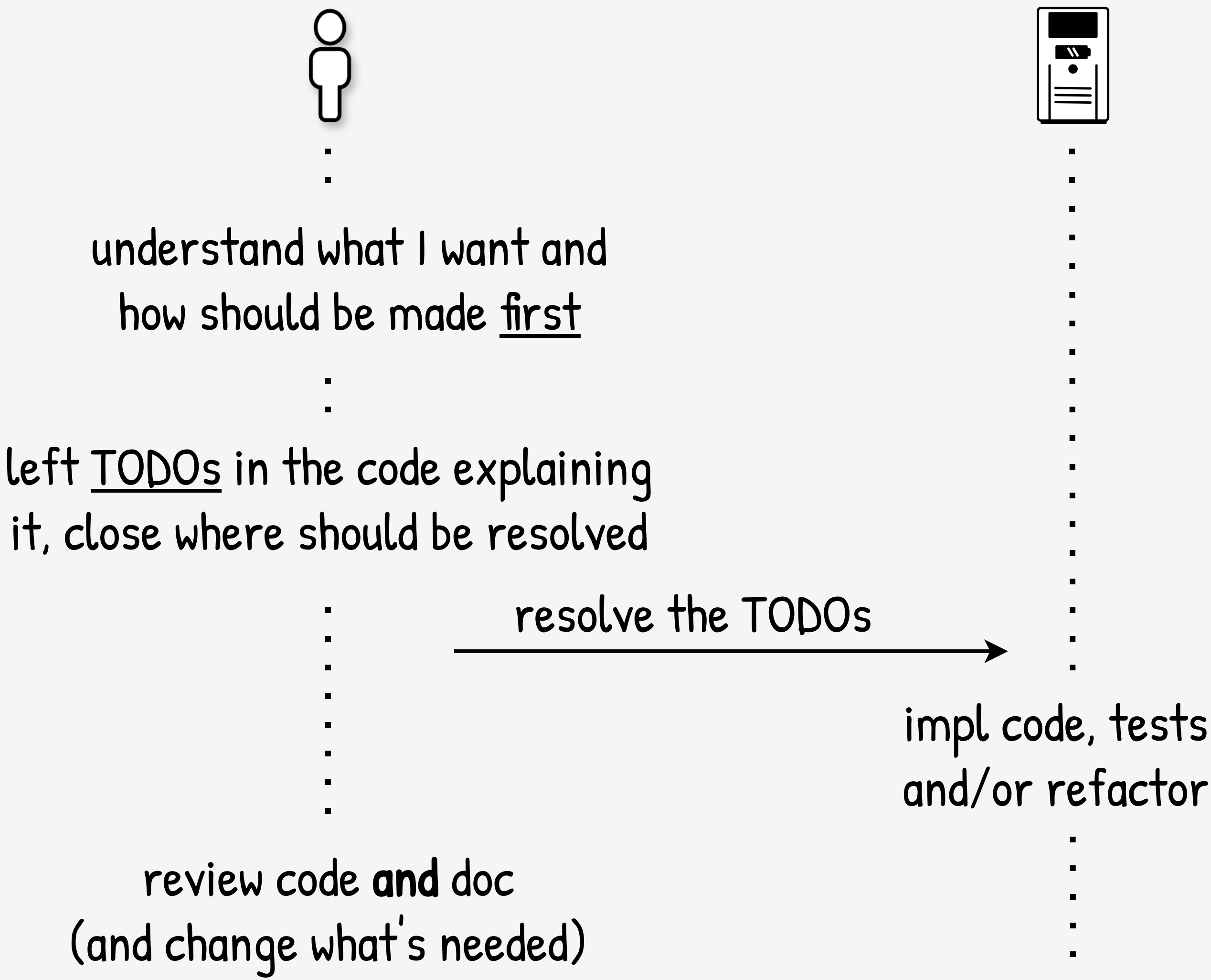

this was the only way
delete everything related with the feature
update design doc
look for inconsistencies, unclear, missing things
review: great
ensure I was understanding
impl missing (but no test, no doc)
code: great
impl code
review code and doc (and change what's needed)

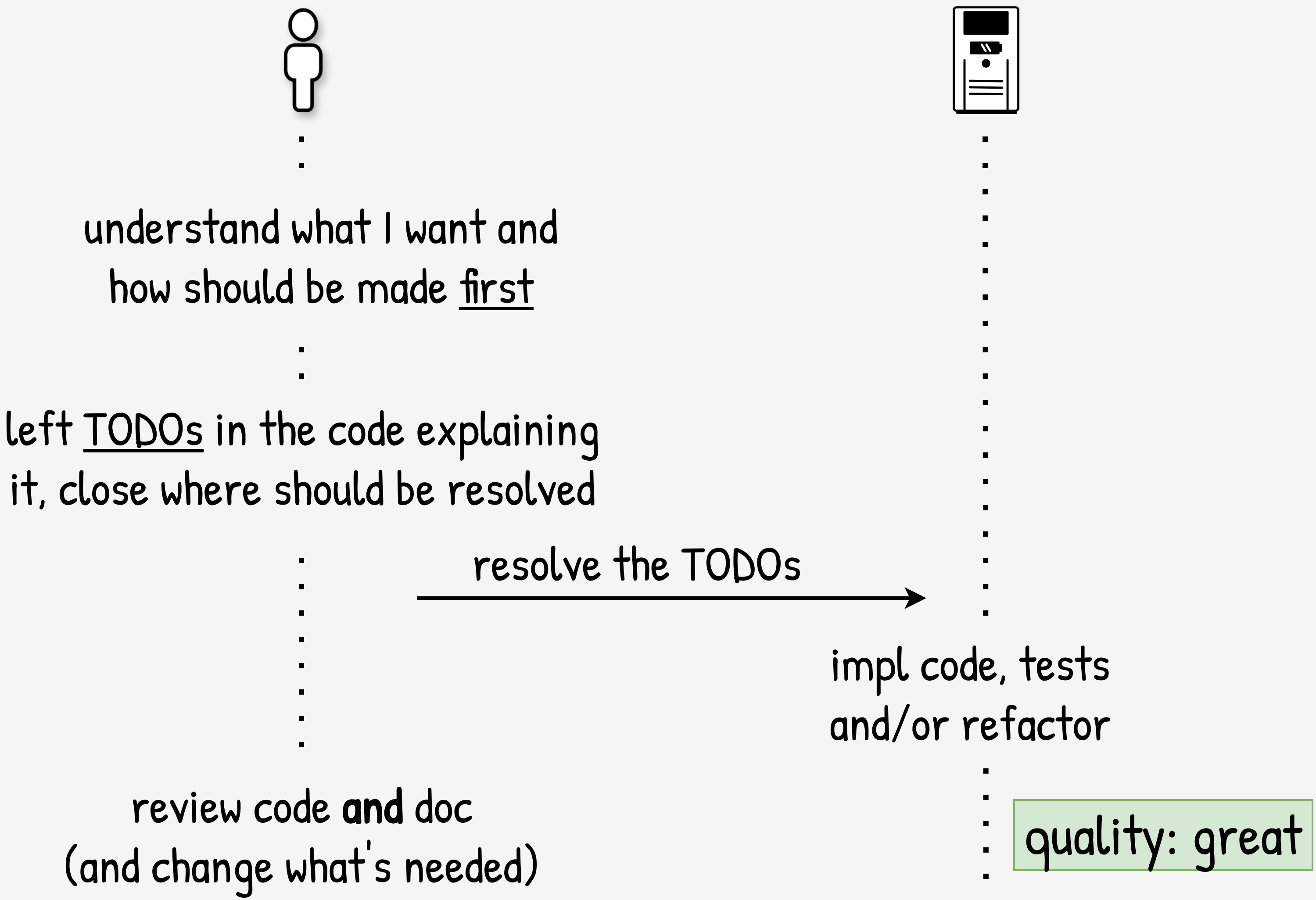

this was the only way
delete everything related with the feature
update design doc
look for inconsistencies, unclear, missing things
review: great
ensure I was understanding
impl missing (but no test, no doc)
code: great
impl code
review code and doc (and change what's needed)
impl few tests
tests: ok
impl tests
review tests (and change what's needed)

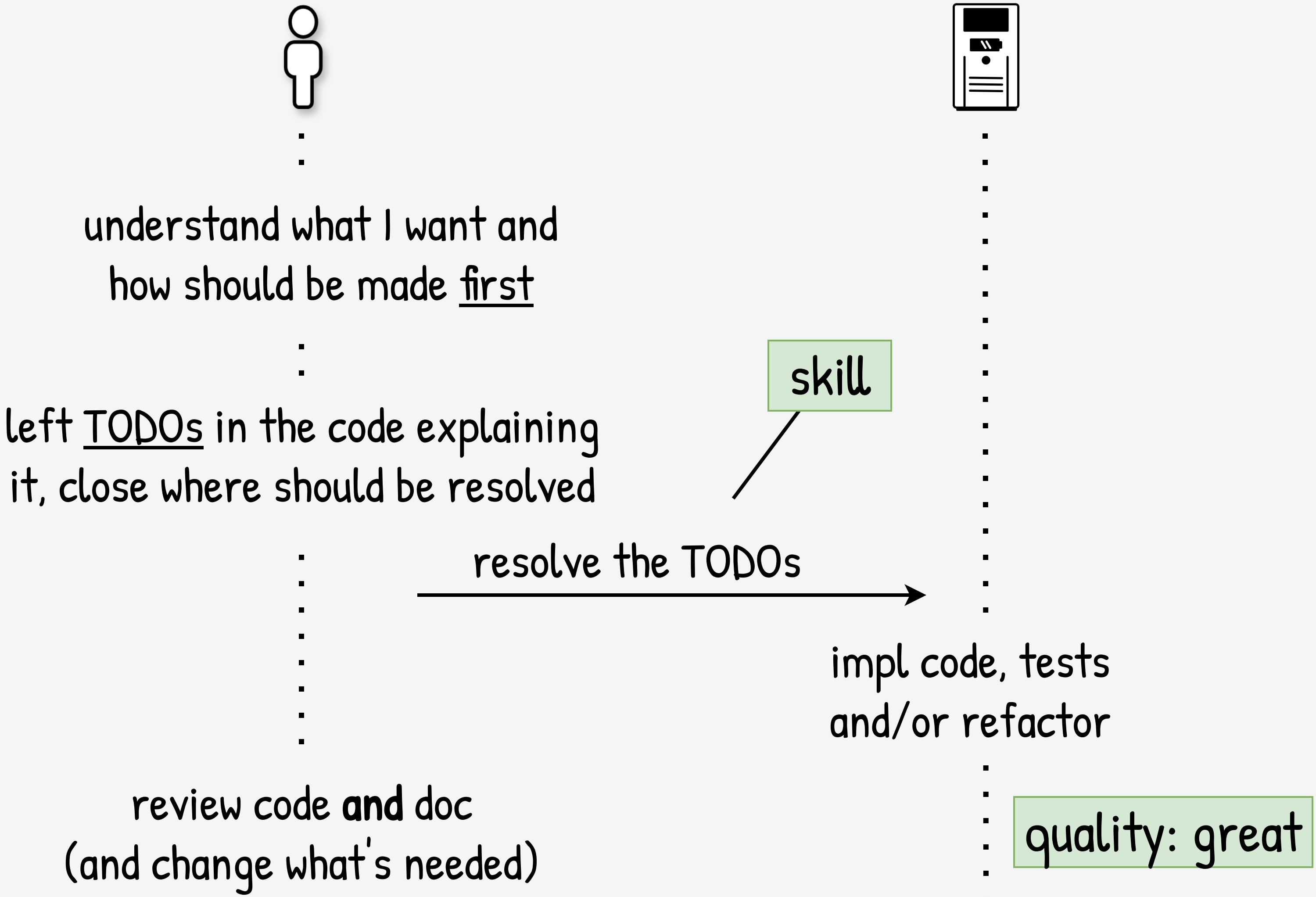

this was the only way
delete everything related with the feature
update design doc
look for inconsistencies, unclear, missing things
review: great
ensure I was understanding
impl missing (but no test, no doc)
code: great
impl code
review code and doc (and change what's needed)
impl few tests
tests: ok
ensure quality
impl tests
review tests (and change what's needed)

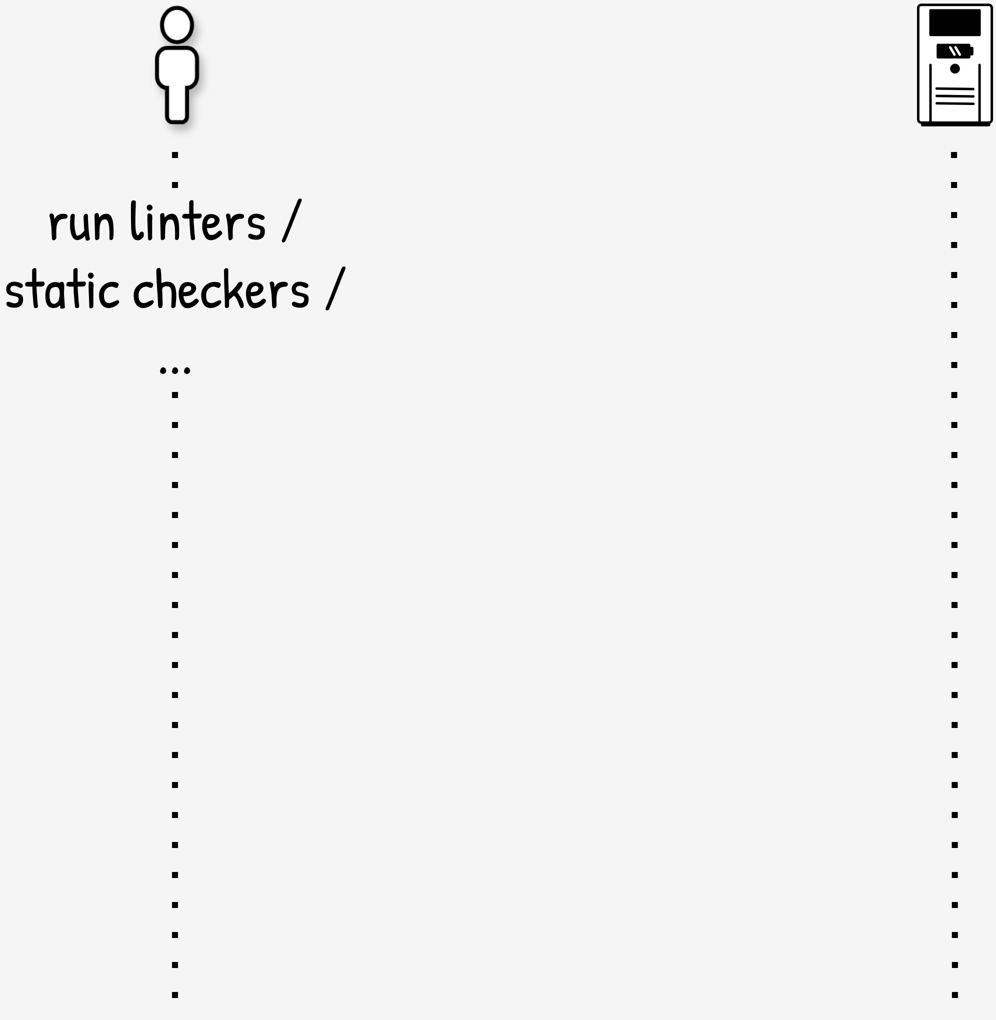
understand what I want and
how should be made first

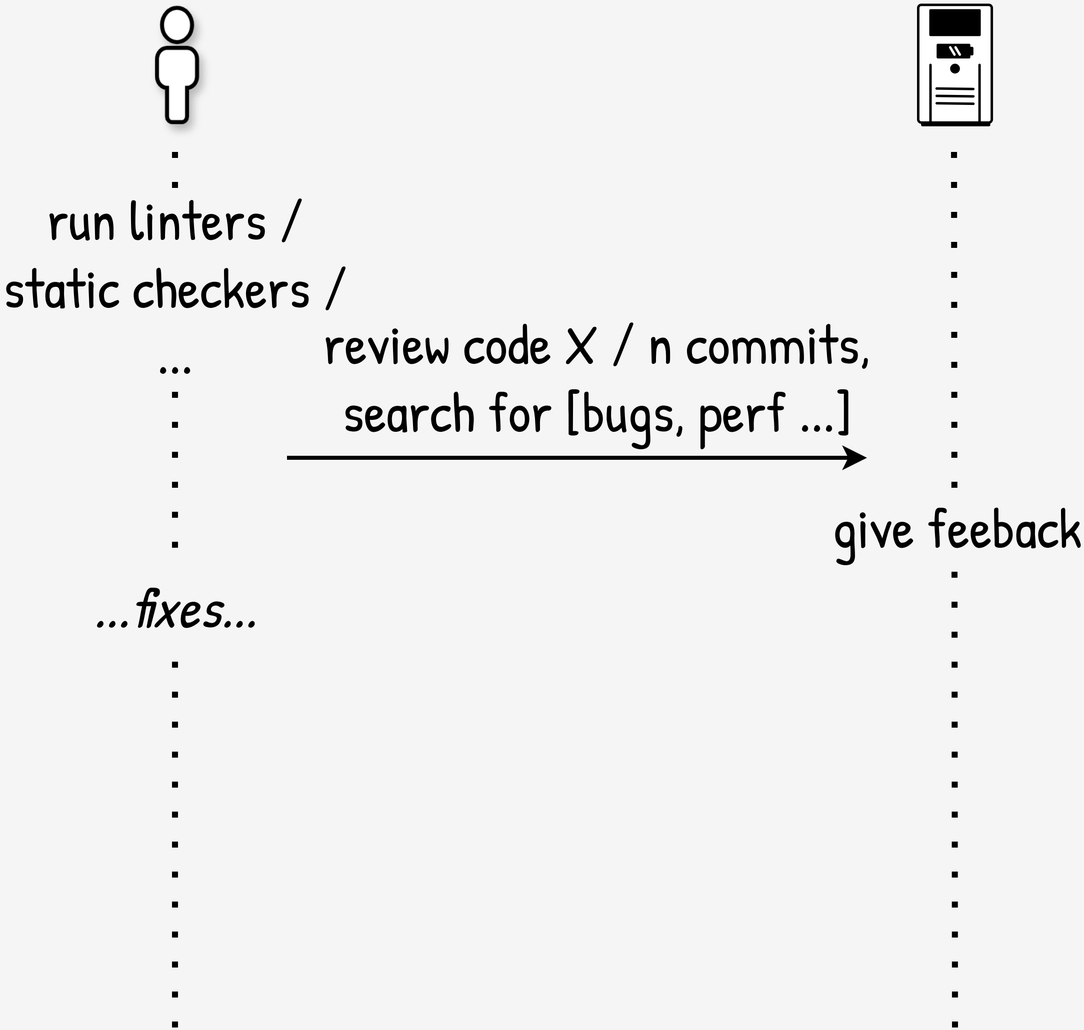
left TODOs in the code explaining
it, close where should be resolved


understand what I want and
how should be made first
left TODOs in the code explaining
it, close where should be resolved
resolve the TODOs
impl code, tests
and/or refactor
review code and doc
(and change what's needed)

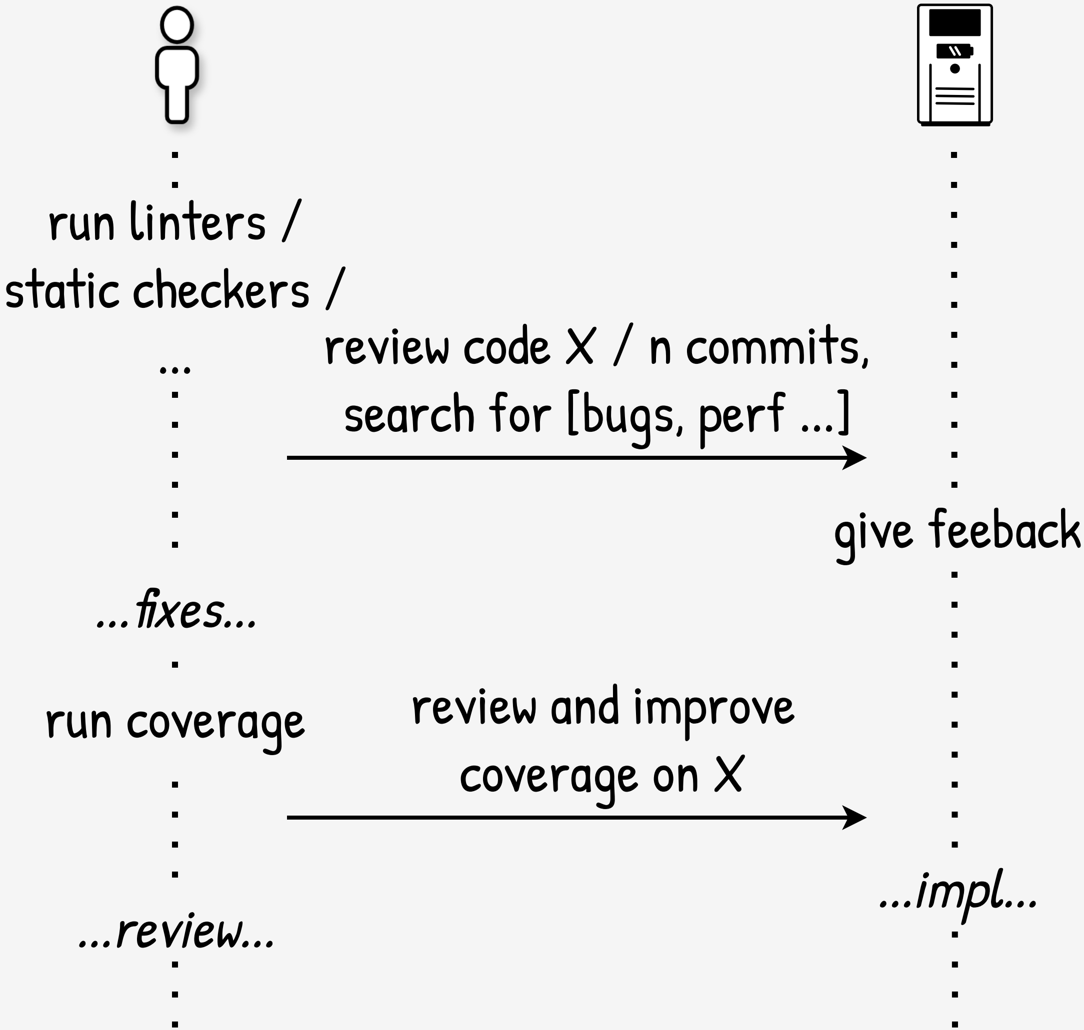

understand what I want and how should be made first
left TODOs in the code explaining it, close where should be resolved
resolve the TODOs
impl code, tests and/or refactor
review code and doc (and change what's needed)
quality: great

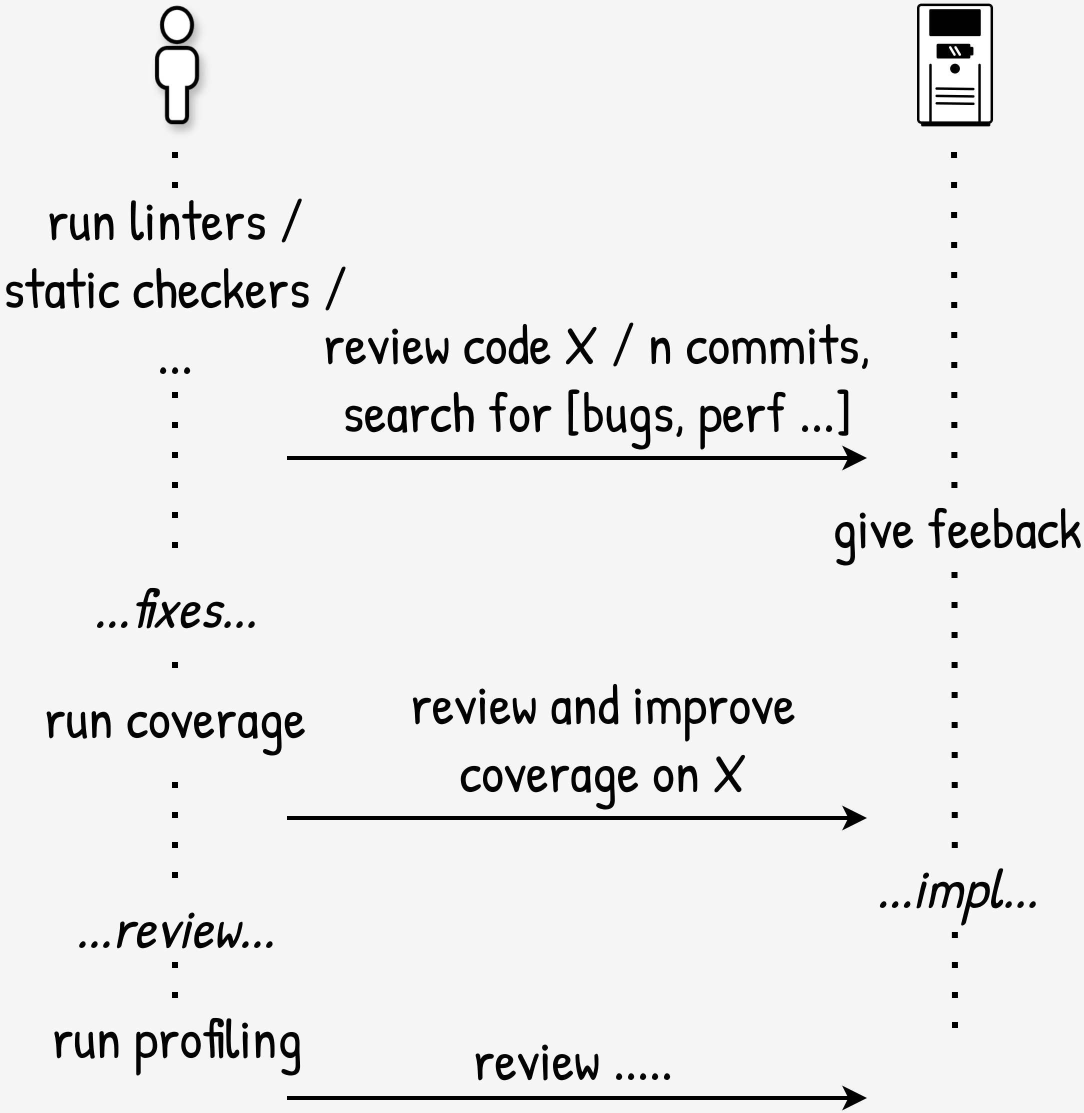

understand what I want and
how should be made first
left TODOs in the code explaining
it, close where should be resolved
skill
resolve the TODOs
impl code, tests
and/or refactor
quality: great
review code and doc
(and change what's needed)

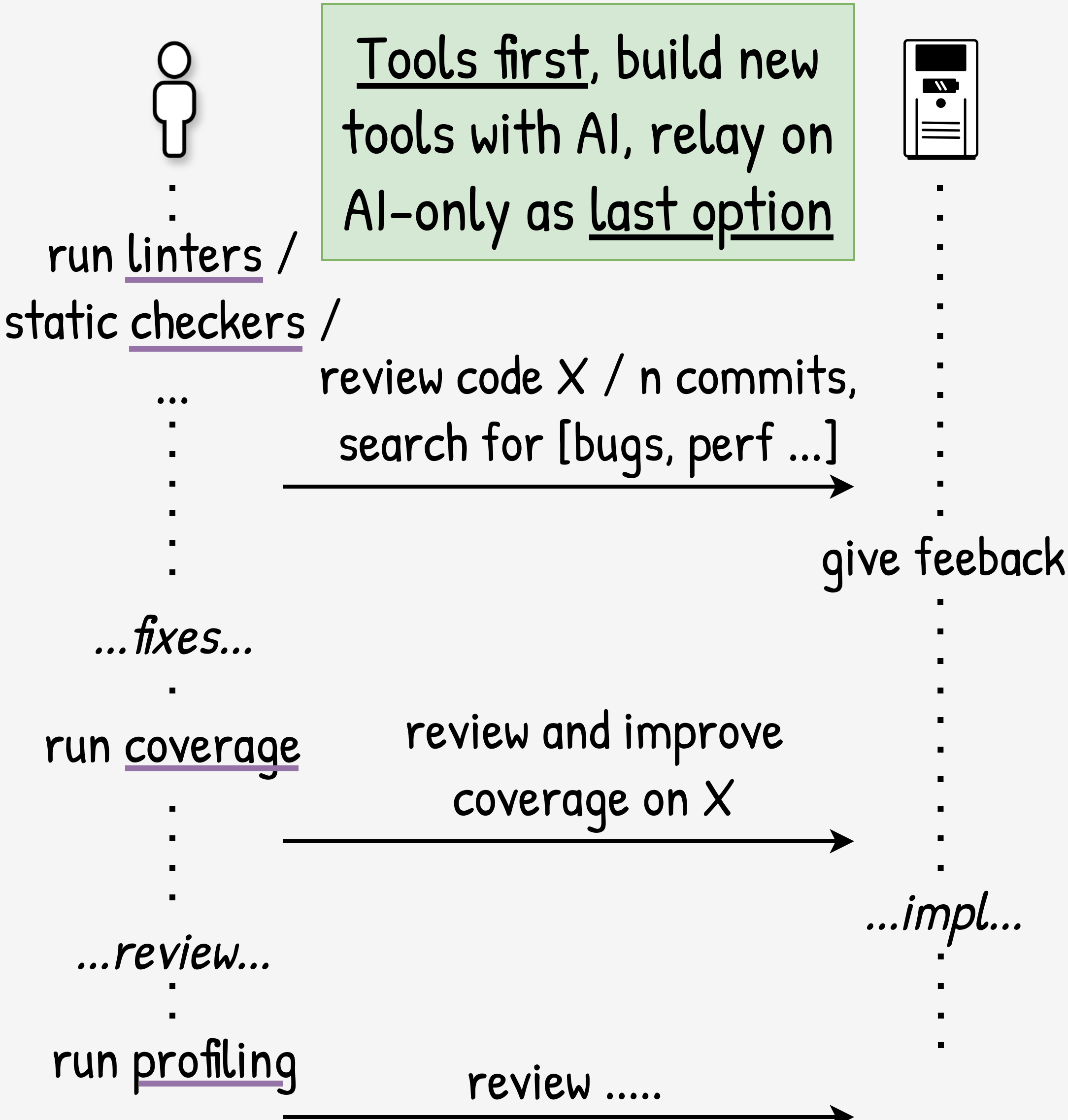

run linters /
static checkers /
...



run linters /
static checkers /
...
review code X / n commits,
search for [bugs, perf ...]
give feeback
...fixes...

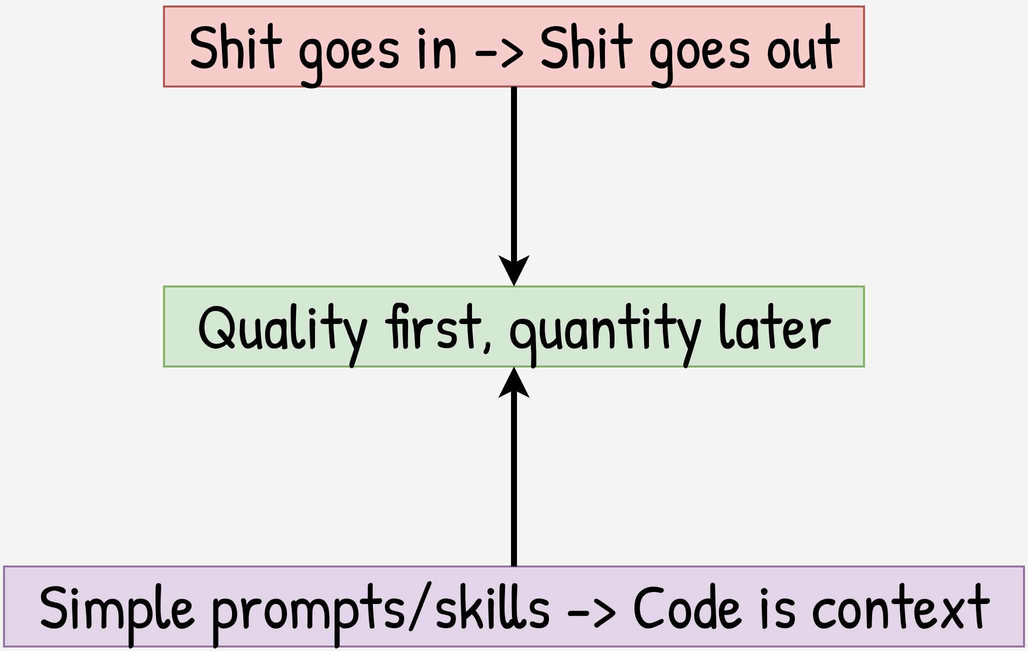

run linters /
static checkers /
...
review code X / n commits,
search for [bugs, perf ...]
give feeback
...fixes...
run coverage
review and improve
coverage on X
...impl...
...review...


run linters /
static checkers /
...
review code X / n commits,
search for [bugs, perf ...]
give feeback
...fixes...
run coverage
review and improve
coverage on X
...impl...
...review...
run profiling
review .....


Tools first, build new tools with AI, relay on AI-only as last option
run linters /
static checkers /
...
review code X / n commits, search for [bugs, perf ...]
give feeback
...fixes...
run coverage
review and improve coverage on X
...impl...
...review...
run profiling
review .....

Shit goes in -> Shit goes out

Simple prompts/skills -> Code is context


Shit goes in -> Shit goes out
Quality first, quantity later
Simple prompts/skills -> Code is context